

1. Discussion

The analytical pipeline applied to 21 years of monitoring data across 88 wells yields a consistent picture of a coastal dune aquifer whose water balance is governed by the interplay of rainfall recharge, atmospheric evaporative demand and head-dependent drainage with land cover operating as a surface boundary condition on a sand body whose effective storage, once interception is corrected, is broadly uniform. Five hydrogeological clusters emerge from the data, each with distinct mechanistic coefficients but broadly convergent specific yield once canopy interception is corrected — convergence carried by C4 and the continuous yield surface, with the clip-constrained C5 estimate only weakly corroborative. The ecological viability of the dune slack system depends on a narrow 37 cm window of summer minimum water table depth (Curreli et al., 2013; Section 4.7.4), and summer minimum depth — not winter rainfall — is the dominant predictor of winter flooding potential across the western and forest clusters (Table 12). This seasonal asymmetry is the organizing principle of the discussion that follows.

The cluster structure links groundwater behaviour to geology and land cover, establishing the physical contrasts that determine how each part of the aquifer responds to climate forcing (Section 5.1). The SSM framework and its benchmarking against the traditional linear model demonstrate that head-dependent drainage memory is essential for long-range prediction (Section 5.2), and the water balance decomposition shows how recharge, atmospheric draw and drainage partition differently across the five clusters under the displacement formulation (Section 5.3). These foundations set the context for evaluating the two management interventions in the monitoring record.

Dune scraping produced measurable improvements in summer minimum depth at CEH36, with a benefit that persisted through the post-felling period; the durability of these gains is set by the background climate trajectory and by interactions with adjacent interventions (Section 5.4). The December 2017 clearfell produced a significant mean water table recovery against the Forest control at the Impact well (+120 mm; Section 5.5). The apparent response at the surrounding Edge wells did not survive a balanced analysis. The recovery is concentrated in the mean monthly signal: summer minima at the felled wells tracked the unfelled Forest controls, consistent with the canopy acting simultaneously as a hydrological sink (recharge suppression through interception) and as a microclimatic buffer on atmospheric demand at the water table (Section 5.6). The spatial framework and BACI record together indicate that forest management perturbations propagate south-westward toward Caernarfon Bay rather than south-eastward into the C1 and C2 clusters (Sections 5.7, 5.8).

The discussion brings these threads together into a management synthesis (Section 5.9): the operational domain for direct hydrological intervention, the role of canopy management in

retaining the plantation's buffering function while partially recovering the recharge penalty, and the climate trajectory that sets the binding constraint on all outcomes.

1.1 Linking Groundwater Behaviour to Geology and Land Cover

The analytical framework rests on two foundational datasets: a 95-year climate record from RAF Valley and a 21-year dipwell network spanning 88 wells across the full geomorphological range of the dune system. The climate record establishes that Newborough Warren operates under a persistent annual surplus (mean $P/PET = 1.36$; Table 1) in which inter-annual water balance variability is driven almost entirely by precipitation rather than by evaporative demand — PET shows strong seasonality but negligible inter-annual trend (Section 4.1.1, Figure 4). This surplus is not seasonally uniform: winter rainfall exceeds PET by a factor of three or more, while summer PET exceeds rainfall, concentrating atmospheric demand on the aquifer during the period of lowest recharge input. The seasonal opposition is the primary climatic driver of dune slack hydrology, and the post-2013 period has run substantially warmer than the pre-2013 summer temperature baseline (Section 4.1.1, Figure 5), intensifying the summer deficit. The monitoring network itself is unusually dense for a dune site — 66 reference wells with a median record length of 15.9 years, supplemented by 22 extended-network wells — providing the spatial resolution needed to distinguish cluster-scale responses from site-wide climate forcing. The network mean water level explains approximately half the variance from cumulative climate forcing alone (Figure 4), with the residual variance reflecting the spatial heterogeneity, boundary subsidies and land cover effects that the subsequent analyses decompose.

The hierarchical clustering solution (Section 3.2; Section 4.2, Figures 7 and 8) identifies five spatially coherent and physically interpretable hydrogeological units that collectively explain the principal axes of groundwater variability at Newborough Warren. The first-order contrast between the responsive eastern open dune (C1, C2) and the buffered Western Residual (C3) reflects a well-documented difference in sub-surface stratigraphy, which the mechanistic coefficients resolve as a gradient rather than a hard binary. Shallower sand over till and estuarine deposits in the east constrains storage and drives rapid, flashy responses — fastest at the lake-coupled C1 Lake Edge, and at the larger, more representative C2 Dune archetype still rainfall-responsive but with slower drainage and longer turnover — while a deeper sand column toward the south-west supports the more buffered C3 Western Residual (Stratford et al., 2007; Grootjans et al., 2004). Whether this contrast reflects two distinct substrate bodies or a single estuarine-clay base that crops out on the higher eastern and southern ground and dips toward Caernarfon Bay is examined in Section 5.1.1. The LCSC values (Table 3) quantify this physical contrast: the Eastern Block clusters show low LCSC values reflecting their constrained storage, the Western Residual is intermediate, and the Forest clusters show the highest values reflecting the combined effect of their inferred deep sand substrate and canopy interception.

LCSC is closely related to specific yield but is not identical to it: the SSM's β_1 coefficient absorbs canopy interception losses alongside true storage response, so that at the forested clusters the elevated LCSC partly reflects the 24% rainfall interception by the Corsican pine canopy (Freeman, 2008) rather than a difference in aquifer drainability. After applying canopy interception correction, the WTF-derived specific yield for C4 falls within the range of the open dune clusters and C5's sits at its upper edge (clip-constrained, weakly corroborative; Section 4.2.4, Table 4c), indicating that the forest clusters' elevated LCSC reflects canopy boundary conditions at the surface rather than differences in subsurface storage architecture. The LCSC framework translates a qualitative geological description into a management-relevant metric expressing the precipitation depth required to achieve a unit rise in the water table: open-dune wells respond to rainfall more efficiently than forested wells, the latter reflecting canopy interception rather than storage capacity (Table 3).

The plantation-covered forest clusters (C4 and C5) are inferred to share the deep sandy substrate of the Western Block, yet segregate into distinct hydrogeological units (Figure 14a). Land cover therefore modifies the hydrological signature of the underlying geology sufficiently to produce a distinct cluster signature. C4 occupies the highest mean water table elevation in the network yet shows the weakest drainage decay coefficient in the network (Table 3), while C5 shows an intermediate drainage rate consistent with its position between the forest interior and the coastal margin. C4's combination of high topographic position and slow drainage reflects two reinforcing mechanisms: canopy interception dampens the recharge pulse so that the water Table 16 never rises sufficiently above its suppressed equilibrium to generate the strong head-dependent drainage response seen in the open dune clusters, and weak lateral discharge produces the within-cluster heterogeneity in β_3 (Section 4.9.5), consistent with the irregular bedrock base that outcrops within the forest and on which C4 sits; its detailed geometry and depth beneath the cluster are not constrained without coring. The displacement formulation (Section 3.4.1), which references drainage to a datum 3.7 m below the pipe top rather than to the ground surface, resolves C4's β_3 to a clearly significant value (Table 3), indicating that head-dependent drainage operates at all five clusters when the hydraulic reference level is correctly specified.

The cluster-level physical reality is independently cross-validated by the state-space model parameters (Section 4.2.2, Table 3). The Western Block's lower recharge sensitivity reflects the deep unsaturated sand buffer that attenuates the rainfall-to-water-table transfer. The Eastern Block's higher β_1 and β_3 values are consistent with a fill-and-spill mechanism in which shallow till promotes surface ponding and rapid lateral overflow rather than slow drainage through the aquifer matrix. The structural divide is evident in the winter flooding record: C1 and C2 reached the wet slack flooding threshold in most years, C3 in a small fraction, C4 in none and C5 in only a single year across the monitoring record (Section 4.8.3, Figure 45). The consequences of this asymmetry for summer minimum trajectory and forecasting are developed in Section 5.2; the anomalous C5 decline is discussed in Section 5.7.2.

The implication for conservation management is that Eastern Block slacks respond sharply to individual rainfall events but retain little water between events, making them sensitive to both exceptional winter rainfall and prolonged summer drought in ways that the buffered Western Residual is not. The post-2018 amplification of C1 seasonal amplitude (Section 4.2.1) is consistent with this sensitivity: the post-2018 period contained two of the three drought summers identified in the record (2018, 2022), producing deeper summer minima at the lake-buffered cluster. Because the entire ecological gradient of dune slack vegetation is compressed into a 37 cm window of summer minimum depth (Curreli et al., 2013; Section 4.7.4), any management intervention must be evaluated primarily on its effect on summer minima rather than on annual-mean water balance metrics.

The cluster structure is independently characterized by the aquifer diagnostic synthesis (Section 4.9.3, Figures 48 and 50), in which three parameters estimated by methods independent of the clustering — drainage decay half-life ($t_{1/2} = \ln(2)/\beta_3$), SSM forecasting gain over the TLM (ΔNSE), and WTF-derived specific yield - place the five clusters in distinct, non-overlapping regions of diagnostic space. C1 Lake Edge occupies the short- $t_{1/2}$, high- ΔNSE , low-Sy corner (shallow, fast-cycling, drainage-dominated), while C4 Main Forest occupies the opposite extreme (long- $t_{1/2}$, low- ΔNSE , intermediate-Sy - sluggish drainage persisting across years). C5 Coastal Forest separates cleanly from C4 on both axes despite sharing the same canopy, consistent with elevation — and the substrate inferred to accompany it — as a major contributor to the diagnostic signature within the forest zone (Section 5.3.1). This model-based characterization of aquifer architecture is consistent with, but derived independently of, the Ward's clustering applied to the raw hydrograph data.

1.1.1 Cluster Classification and Validation

Two diagnostic checks support the aquifer-architecture framing of Section 5.1 with quantitative evidence drawn from the network's own monitoring record (*28_c3_detrend_check.py* and *29_c3_within_variance_check.py*). The first tests the hypothesis that the C3 Western Residual cluster, which behaviourally separates from the C2 Dune cluster on both Ward's clustering and the diagnostic synthesis (Section 4.9.3), might in fact be C2 with a western-margin coastal-erosion drift superimposed. For each C3 well with available coast-distance metadata, the monthly hydrograph was de-trended against the network-scale coastal-retreat gradient model (Section 4.8.2; forest-free linear-capped fit, $\delta_0 = -29.0 \text{ mm yr}^{-1}$, $L = 894 \text{ m}$) and re-classified against the un-de-trended cluster centroids using the same correlation-distance metric that Ward's used to construct the clusters. Of nineteen testable C3 wells, none migrated from a C3 best-match to a C2 best-match as a result of the procedure. The result is robust across five sensitivity variants — full-fit versus forest-free δ_0 , three values of L , and summer-only versus monthly-uniform de-trending — and is corroborated by a C2 sanity-check retention of 24 of 24 wells, confirming that the de-trending procedure does not contaminate hydrographs of wells that the gradient model attributes negligible drift to. The C2/C3 behavioural distinction therefore cannot be

reproduced by removing a linear coastal-erosion drift, consistent with a genuine aquifer-architecture control — whether two distinct substrate bodies or a single estuarine-clay base dipping toward the bay (Section 5.1, and below) — rather than a coastal-drift artefact (shallow sand over till in the east, deeper sand in the south-west; Stratford et al., 2007; Grootjans et al., 2004).

The second check examines within-C3 heterogeneity. The per-well coefficient maps (Section 4.9.2) visibly show structured variation in β_1 , β_2 , β_3 and the drainage decay half-life $t_{1/2}$ across the C3 wells; the question is whether this within-cluster variance follows a coherent spatial axis. A panel regression of nine per-well behavioural metrics against five spatial and hydrogeological predictors (the Section 4.8.2 exponential coastal predictor, distance to CEH36, distance to the forest polygon edge, ground elevation, and depth to water) returns $R^2 = 0.81$ for β_1 recharge, 0.74 for β_3 drainage, and 0.70 for the drainage decay half-life $t_{1/2}$. The exponential coastal predictor's regression coefficient against per-well slopes lands at +1.0, an independent face-value validation of the Section 4.8.2 functional form. The strongest unique predictor across all four SSM coefficients and the storage–drainage index is distance to CEH36 — a hydrogeological axis within C3 anchored near the south-western interior of the warren, distinct from the network's coastal-margin gradient. A sensitivity check at $k = 6$ confirms that this within-C3 structure does not warrant a finer cluster partition: at $k = 6$, Ward's correlation-distance clustering keeps all 21 C3 wells together as a single cluster and instead splits C2 into a seasonal-amplitude-distinguished pair. C3 is therefore "structured but not strongly enough to split" — the within-cluster hydrogeological axis is real and informative for spatial interpolation and per-well diagnostics, but does not destabilize the $k = 5$ partition the report adopts throughout.

These two strands — a genuine, non-coastal C2/C3 distinction, and a coherent within-C3 gradient along the south-west to north-east axis (recharge sensitivity β_1 and drainage rate β_3 rising inland, $r = +0.76$ and $+0.58$; WTF-derived specific yield falling inland from approximately 0.41 at the south-west margin to 0.25 at the north-east, $r = -0.83$, $p < 0.001$) — admit a parsimonious reading. A single estuarine-clay base that crops out on the higher eastern and southern ground, where a thin overlying sand column produces the flashy Dune response, and dips beneath the surface toward Caernarfon Bay, where a thickening sand column produces the buffered C3 response, would make C2 and C3 successive positions on one dipping system rather than two discrete blocks — C3 grading toward the open dune at its inland edge and toward the forest and coast at its margins, consistent with its diffuse affinity signature (Section 4.3). The geometry is interpretation, not observation: the clay base is not cored, and coastal position, plantation proximity and the inferred clay dip covary along the same axis, so two distinct depositional bodies fit the same coefficient evidence equally well. A cored or geophysical south-west to north-east transect carrying the Caernarfon Bay coastal-retreat rate as a covariate would be required to separate them (Section 5.7.3).

The gradational nature of the aquifer between these structural units is shown by the Pearson affinity analysis (Section 3.3; Section 4.3). Of the 66 reference wells, the great majority confirmed their primary cluster membership; three Spy wells showed strongest temporal affinity with a neighbouring cluster at borderline correlation deltas, consistent with gradational boundaries rather than misclassification. The Multi-Cluster Affinity flag was met at a substantial fraction of wells, reflecting the high mutual correlation among the five centroids in a contiguous, highly transmissive sand aquifer (Rao and Srinivas, 2006). The spatial concentration of MCA flags along the C1/C2/C3 transitional zone, the C2/C3/C5 boundary where the Dune, Western Residual and Coastal Forest signatures converge, and the C3/C4/C5 forest–dune boundary (Figure 14a) identifies the areas where geological and vegetative controls overlap. Hydrogeologically, these are candidate locations for rapid state change in response to sustained conservation management or climatic intervention (Grootjans et al., 2004).

The persistence of Forest cluster affinity at the clearfell treatment wells (Section 4.3) admits at least three interpretations: residual suppression by the surrounding intact canopy extending beyond the felling boundary; incomplete hydrological recovery within the felled tabfootprint itself; or, most fundamentally, that the Forest cluster signature at these wells reflects high topographic elevation and its inferred deep sandy substrate rather than canopy influence alone — in which case convergence toward open dune behaviour may not be achievable regardless of management intervention. Distinguishing between these mechanisms requires continued monitoring beyond the current record.

1.2 Boundary Conditions, Model Limitations and Forecasting

The SSM is a lumped mass-balance model fitted independently at each well. It does not resolve lateral groundwater flow, and its three-term structure — recharge ($\beta_1 \cdot P$), atmospheric draw ($\beta_2 \cdot \text{PET}$) and head-dependent drainage ($\beta_3 \cdot h_{\text{disp}}$) — necessarily omits any flux that does not covary with rainfall or PET at monthly resolution. Where the three-term balance does not close, the residual appears as a structural mismatch between predicted and observed water levels.

1.2.1 Water Balance Residuals as a Model-Adequacy Diagnostic

The water balance residual field (Section 4.9.7, Figure 58) makes this mismatch spatially explicit, and its result is largely negative: across the reference network the three-term balance closes without requiring an additional flux. No well exceeds +0.02 m/month, the eight positive residuals all fall below +0.005 m/month, and all five cluster medians are negative (Section 4.9.7). Residual magnitude is uncorrelated with position on either axis (Spearman $\rho = +0.099$ against Easting, $p = 0.43$; $\rho = +0.111$ against Northing, $p = 0.37$). The signed residual carries a weak west-to-east gradient short of conventional significance ($\rho =$

-0.226 against Easting, $p = 0.07$), while the ridge-to-dune axis, on which a ridge-derived input would express itself, remains null ($\rho = -0.171$ against Northing, $p = 0.17$); the Easting gradient is reported for completeness rather than as an established structure.

The field is therefore a diagnostic of where the SSM's three terms are sufficient rather than a map of lateral inflow. It should not be read as the latter in any case, because the residual also absorbs nonlinear soil-moisture storage dynamics, β coefficient sensitivity and any systematic bias in the Thornthwaite PET estimate. On the present evidence no unmeasured boundary flux need be invoked to close the balance at any cluster, and the independent water balance of Script 16 agrees with the field to within ± 5.8 mm/month at every cluster.

The independently computed Darcy flow vectors derived from the mean head gradient field (Section 4.9.6, Figure 57) indicate a radial flux diverging from the north-western high point, southward and south-westward in the western half of the site and south-eastward in the eastern half. They describe the site's flow geometry and are retained on that basis; with no ridge-proximal residual signal, they do not bear on the residual field.

Two independent diagnostic tests on the SSM time-series residuals return null results and are reported in full in Supplementary Notes S5 and S6: a cross-correlation lag analysis testing for distance-dependent transport from the ridge (Spearman $\rho = -0.005$, $p = 0.97$, $n = 50$) and a seasonal climatology analysis testing for unmodelled summer evaporative demand (network mean $r = -0.026$ against sunshine hours). Note that these operate on the monthly SSM regression residual $\varepsilon(t)$, a different quantity from the scalar water-balance closure residual mapped in Figure 58.

The seasonal residual diagnostic (Script 24) provides a further constraint. The dominant phase of the SSM residual peaks in autumn through early spring, with no wells of the 63 analysed peaking in summer. This is not the signature of unmodelled summer evapotranspiration — if Thornthwaite PET were systematically underestimating summer losses, the residual would peak in summer months.

1.2.2 Displacement Formulation and Depth-Dependent PET

The displacement formulation (Section 3.4.1) — which references drainage to a datum 3.7 m below the pipe top rather than to the ground surface — captures head-dependent drainage at all five clusters, with β_3 significant at $p < 0.01$ or better throughout the network (Table 3). Under the pre-displacement specification, C4's drainage term was borderline; the displacement formulation resolves this by correctly specifying the hydraulic reference level against which drainage operates (Section 5.1). The per-well datum sensitivity analysis (Section 4.9.1, Figure 47a) shows that the uniform 3.7 m datum produces negligible R^2 loss relative to per-well optimized datums across most of the network, with meaningful gains limited to C1 Lake Edge wells where the effective drainage base is shallowest.

A depth-dependent extension of the SSM, in which the β_2 coefficient declines exponentially with water table depth according to $\beta_2(d) = \beta_2 \cdot \exp(-\lambda d)$, was evaluated at cluster level (Section 3.7.1). The depth-dependent formulation improves iterative NSE at all five clusters ($\Delta\text{NSE} +0.028$ to $+0.182$), consistent with capillary connectivity between the water table and root zone diminishing with depth — a real physical effect that the standard SSM's fixed β_2 absorbs as a depth-averaged effective parameter. The improvement is strongest at C1 Lake Edge ($\lambda = 1.90 \text{ m}^{-1}$, $\Delta\text{NSE} = +0.182$), where shallow water-table evaporation near the lake is the dominant ET pathway, and weakest at C4 Main Forest ($\lambda = 0.15 \text{ m}^{-1}$), although C4's narrow depth range means even weak coupling produces a measurable NSE gain ($+0.104$). The standard SSM is retained as the primary operational model because its fixed β_2 preserves the closed-form threshold equations and scenario algebra on which the operational outputs depend. The mechanistic implications — particularly the finding that C4 shows the weakest depth coupling despite having the highest β_2 — are discussed in Section 5.6.1.

1.2.3 The C4 Main Forest Drainage Coefficient: Identifiability and Sensitivity

The drainage coefficient $-\beta_3$ is markedly low at the C4 Main Forest cluster. The cluster centroid returns $-\beta_3 = 0.020 \text{ month}^{-1}$, the lowest in the network (Section 4.2.2), and one constituent well (CEH14) returns a physically inadmissible negative value. A low coefficient of this kind can arise in two quite different ways: as a genuine hydrogeological property, or as an artefact of the atmospheric-draw term absorbing variance the drainage term would otherwise hold. Because the distinction bears on both the C4 water balance and the interpretation of the cluster's high β_2 , the fit was tested directly for that degeneracy (Script 30).

The diagnostic does not support it. The drainage predictor in the SSM is the displacement of the water table above the 3.7 m drainage datum at the start of each month (Section 3.3), and the question is whether that predictor is separable from PET. At C4 it is the most cleanly separated in the network: the variance inflation factor on the displacement term is 1.11 and the correlation between PET and displacement is -0.05 , against 1.76 and -0.60 at C1, 1.37 and -0.42 at C2, and 1.25 and -0.31 at C3. Nor is the term starved of signal — C4 carries the network's largest displacement variation (standard deviation 0.45 m). The coefficient is statistically significant at the centroid the report cites ($p = 0.003$). The atmospheric-draw and drainage coefficients are separately identifiable at C4, and the low drainage coefficient is a property of the data rather than an artefact of the fit.

The per-well scatter that first raised the question has a simpler explanation. Six of the nine C4 wells return a non-significant β_3 and one (CEH14) returns a negative value, but the median per-well variance inflation factor is 1.18, again the lowest in the network: the scatter reflects limited statistical power in resolving a coefficient of order 0.02 month^{-1} from a hundred-month record, not a β_2 – β_3 trade-off. What trade-off appears is carried almost entirely by one well. The per-well correlation between β_2 and β_3 is -0.68 across all nine

wells, -0.23 with CEH14 removed, and -0.14 with CEH13 and CEH14 both removed — the two wells the pipeline already sets aside on independent grounds. The three directly identified wells (NW10, CEH20, CEH30) span 0.024 – 0.043 month $^{-1}$, and the measured Corsican-pine wells centre on 0.025 – 0.030 .

This has a direct — and reassuring — consequence for the cluster water balance (Section 4.2.3, Figure 11). Recomputing the C4 partition across the plausible range, from the unconstrained centroid (0.020) through the identified-well values (0.025 – 0.030) to an upper bounding value of 0.058 month $^{-1}$ — the open-dune drainage rate adjusted for the forest tree-and-substrate effect inferred from the inland C5–C3 contrast, retained here only as a bound — leaves the qualitative result unchanged across the entire identified range: the forest-interior loss budget remains led by atmospheric draw. The drainage share rises only modestly, from 24% at the unconstrained value to roughly 31% at the identified-well mean and 34% at its upper bound, and crosses into a drainage-led partition (56%) only at the bounding value. The water-balance closure provides an independent check that favours the identified range over either extreme: the cluster closes most tightly (residual essentially zero) at $\beta_3 \approx 0.026$, and the residual grows in both directions — to $+0.8\%$ of the loss budget at the unconstrained value and to -4% at the bound, the largest closure residual in the network. The identifiability diagnostic, the identified per-well coefficients and the water-balance closure thus converge on $\beta_3 \approx 0.026$ ($t_R \approx 38$ months): the forest interior drains genuinely more slowly than the open dune (C2 $t_R \approx 14$ months), a real canopy-and-substrate effect. The report retains the unconstrained fit as its primary record; this section establishes that the principal water-balance conclusion is robust across the full plausible range.

One consequence follows directly for the atmospheric-draw coefficient. Because β_3 is cleanly identified at C4, the cluster's high β_2 cannot be read as a collapsed drainage term reappearing in the evaporative one: C4's β_2 of 2.58 is a real feature of the cluster, standing well above the open-dune clusters (C2, 1.76; C3, 1.83) rather than resembling them. That is not, however, the same as saying that more water evaporates beneath the plantation. β_2 measures the water-table response per unit of atmospheric demand, not the evaporative flux itself: the same loss produces a larger head change where less drainable storage is available to absorb it, so the coefficient carries the substrate as well as the demand. The two forest clusters make the point directly. C4 and C5 carry broadly the same Corsican pine canopy, yet between them they span almost the whole network range of β_2 — C4 the highest at 2.58, C5 among the lowest at 1.31, below both open-dune clusters. A canopy effect common to both cannot produce a two-fold contrast between them. What differs is position and substrate: C4 on the ridge flank over a thin, low-storage sand column, C5 on deeper coastal ground whose higher WTF-derived S_y converts a given evaporative flux into a smaller head change (Section 5.6.1). The report therefore treats the unconstrained C4 β_2 as real, and reads it as a storage-mediated head response rather than as evidence of elevated evaporation beneath the canopy.

CEH14 is treated separately. On the ridge flank, it is the single well where the SSM fails outright (iterative NSE = -3.21 , the worst fit in the network; Section 4.4): the displacement formulation does not capture the bedrock-ridge boundary condition, and its negative $-\beta_3$ is a symptom of that failure rather than a measurement. It is excluded from the drainage decay half-life map (Section 4.9.3) and the per-well mechanistic interpretation on that ground. The forest-wide structure of the positive water-balance residuals — which extend beyond C4 into the Coastal Forest and the forest-margin wells of adjacent clusters — is a distinct, physical signal and is discussed under boundary subsidies in Section 5.2.1; it is not a drainage-coefficient artefact, and a β_3 adjustment would not account for it.

1.2.4 Predictive Skill and Operational Forecasting

The physical distinction between the SSM and the traditional linear model (TLM) is most directly revealed by long-range iterative simulation (Section 3.4.2; Section 4.4). In one-step diagnostic mode both models perform almost identically (median R^2 0.91 and 0.92 respectively; Table 5), because each prediction is corrected with the observed water level from the previous month. The divergence is substantial under iterative simulation: the TLM achieves a median iterative NSE of 0.18, while the SSM achieves 0.75; the median per-well improvement is $+0.53$ NSE units (Table 5). The SSM returned a positive iterative NSE at 65 of 66 wells compared with 44 of 66 for the TLM (Table 5; Figure 16). The explicit β_3 term in the SSM captures storage decay memory, preventing the cumulative drift that otherwise causes TLM divergence. The Main Forest cluster (C4) shows the smallest improvement ($+0.229$), consistent with two non-exclusive readings: the canopy may smooth the recharge signal into a near-linear climate response that the TLM can handle adequately without a storage decay term, and the structurally suppressed water table at C4 carries less nonlinear storage dynamics for the SSM to capture. The two readings are not distinguishable from the NSE benchmark alone.

This predictive capacity has three practical consequences for conservation management. First, the SSM can generate forward projections of water table trajectory under specified climate scenarios (Section 4.4), enabling managers to anticipate when summer minima may cross the ecological viability thresholds identified by Curreli et al. (2013) rather than relying solely on backward-looking trend extrapolation. The seasonal prediction equations (Table 12) and P_flood threshold equations (Table 15) are algebraic inversions of the SSM's fitted coefficients — they are not separate models but direct consequences of the SSM parameterization, and their predictive credibility rests on the SSM's demonstrated iterative skill.

Second, the spatial pattern of SSM improvement delineates the aquifer's structural boundaries without invasive coring (Figure 16). The largest NSE gains concentrate in the C1 Lake Edge sector, where substantial seasonal head variation and head-dependent drainage to

Llyn Rhos-Ddu make the storage decay term essential. The smallest gains occur in C4 Main Forest. The improvement map is itself a diagnostic: it identifies which parts of the aquifer are governed by nonlinear storage dynamics and which approximate a linear climate response.

Third, the SSM framework is operationally transferable. The model requires only three fitted parameters per well ($\beta_1, \beta_2, \beta_3$), a uniform drainage datum, and monthly rainfall and temperature from a single climate station. It does not require hydraulic conductivity measurements, aquifer thickness estimates, or lateral flow modelling — the parameter-sparsity constraints that produced calibration difficulties in the Betson et al. (2002) MODFLOW model at this site — where hydraulic conductivity rested on a single tracer test ($K = 6.0$ m/day) and aquifer thickness on four borehole logs across 700 ha (Bristow, 2003; Section 5.7.3). Any coastal dune aquifer with a monitoring network and a nearby climate station can be parameterized in the same framework. The displacement formulation (Section 3.4.1), which references drainage to a fixed datum rather than the ground surface, is the methodological choice that makes this possible: it captures the depth-dependence of the drainage response without requiring site-specific calibration of the drainage geometry.

1.2.5 Seasonal Prediction and the Intervention Window

The demonstration that winter rainfall amount is a poor predictor of peak flood conditions in the Western Residual ($P_{\text{winter}} p = 0.078$; Table 12), whilst antecedent summer minimum depth is dominant ($p = 0.015$), is the most consequential single finding for conservation management. It means that managers cannot reliably forecast slack flooding from seasonal rainfall forecasts alone — what matters is how deep the aquifer drains each summer. The seasonal prediction equations — which regress peak winter water table against antecedent summer minimum depth and cumulative winter rainfall (Section 3.6.2; Table 12) — were fitted with one equation per cluster under the $k = 5$ partition: winter rainfall was significant in the two eastern clusters (Lake Edge $p = 0.002$, Dune $p = 0.004$) but non-significant in the Western Residual ($p = 0.078$), Forest ($p = 0.919$) and Coastal Forest ($p = 0.379$). Antecedent summer minimum depth was the dominant predictor in the three western and forest blocks (Western Residual $p = 0.015$, Forest $p < 0.001$, Coastal Forest $p < 0.001$). The ecological stakes of this finding are clarified by the eco-hydrological thresholds established by Curreli et al. (2013), whose four-year monitoring programme showed that wet slack vegetation (SD15b) requires mean annual maximum water levels above approximately -0.10 m relative to ground surface, and dry slack communities (SD16) persist at winter maxima around -0.25 m. Summer minimum depth sets the precondition for achieving these winter maxima: a water table that has drained deeply through summer must receive proportionally greater winter recharge to recover to flooding height. The P_{flood} equations quantify this precondition directly (Section 3.6.3; Table 15).

The summer minimum trajectory (Section 4.8.2, Figure 45) confirms that this precondition is deteriorating. All five clusters show declining summer minimum trends, with C1 ($p = 0.050$) and C5 ($p = 0.003$) statistically significant, C2 marginal ($p = 0.054$), and C3 ($p = 0.336$) and C4 ($p = 0.398$) non-significant. The post-2013 period is running 0.94°C above the pre-2013 temperature baseline (Section 4.1.1). A bootstrap of the C1 summer-minimum trend (Figure 70; $n = 1000$ year-resampling replicates of the OLS fit) places the median crossing year of the SD16 dry-slack viability threshold at 2022, with a 90% confidence interval of 2017–2042. The wide interval reflects the substantial year-to-year variability in C1 summer minima against a shallow trend slope of -9.7 mm yr^{-1} . The dry summers of 2022 and 2025 have pulled the central estimate forward into the observed record. The cluster is therefore no longer approaching SD16 but oscillating across it: seven of the twenty observed summer minima have exceeded the SD16 depth, including four of the eight most recent years (2018, 2019, 2022, 2025), while wet years (2020, 2021, 2023, 2024) sit above SD16 by 50 to 130 mm. The "crossing" point in the bootstrap is therefore best read as the year by which the underlying trend, averaged over wet-and-dry-year variability, meets the threshold. Recent ecological field surveys provide direct observational validation: Callaghan et al. (2021) found that despite active topographical intervention at Newborough, *Petalophyllum ralfsii* populations remained restricted due to limited water availability, and recorded a complete absence of winter flooding following the severe 2018 summer drought. The intervention window is convergent with projections from other west-coast UK sites: Curreli et al. (2013), applying their community thresholds to the Ainsdale water-table projection of Clarke and Sanitwong Na Ayutthaya (2010), concluded that conditions supporting wet slack vegetation may become non-viable by the 2030s. The intervention window is therefore likely to be measured in one to two decades — and at C1, in years already monitored — rather than the longer timeframes typically assumed in dune restoration planning. One mechanism operates weakly against this trajectory at C1 specifically: the depth-dependent PET diagnostic (Section 4.4; Section 5.2.2) shows C1 to have the strongest depth-coupling in the network ($\lambda = 1.90 \text{ m}^{-1}$, $\Delta\text{NSE} = +0.183$), so as the C1 water table deepens the capillary connection to the evaporating surface weakens and the atmospheric-draw term attenuates. This is a modest negative feedback that slows, but does not reverse, the approach to the SD16 threshold, and it is already embedded in the fixed- β_2 trend the bootstrap projects.

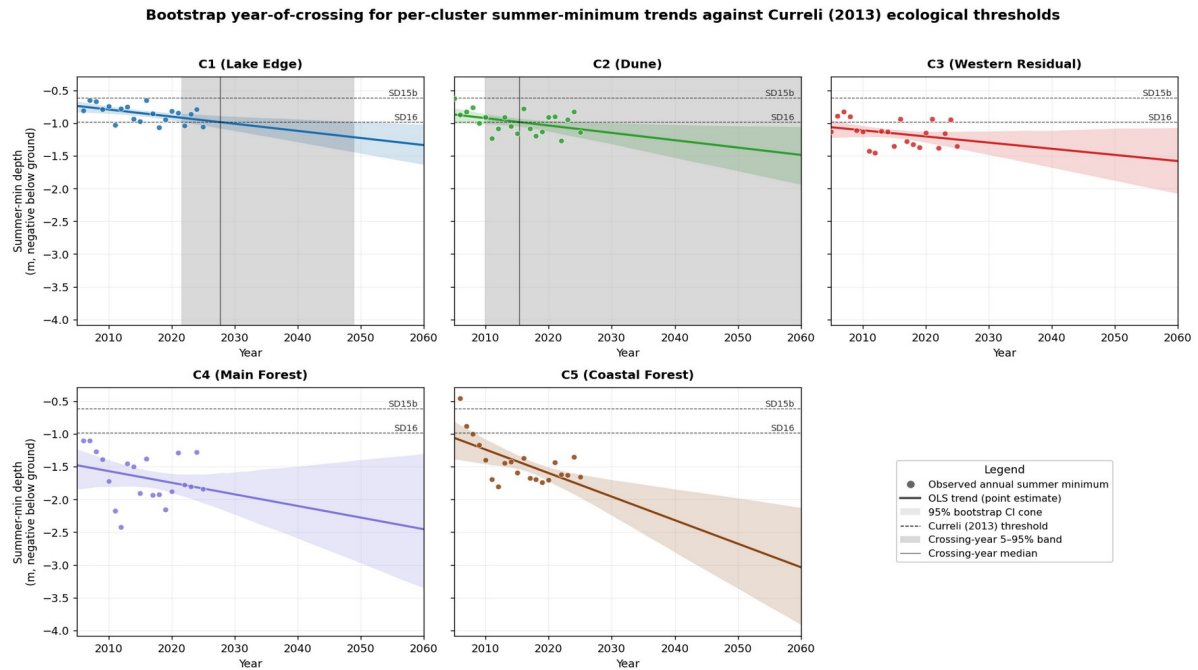


Figure 1.1: Bootstrap year-of-crossing for the five-cluster summer-minimum trends against the Curreli *et al.* (2013) SD15b (0.61 m) and SD16 (0.98 m) ecological thresholds. Each panel shows observed annual summer minima (filled circles), the OLS trend with 95% bootstrap confidence interval (shaded), and the two threshold lines. Vertical grey bands indicate the bootstrap 90% confidence interval for the year at which the trend crosses each threshold; the central line marks the median. The C1 SD16 crossing year has a median of 2022 with CI 2017–2042 under the current data state. C2 has crossed SD16 in the trend sense around 2011 with the CI extending into the pre-monitoring period; C3 lies above SD16 and its non-significant trend reaches the threshold beyond 2080; C4 and C5 sat below SD16 throughout the observed record as the forest-zone clusters do not host slack vegetation. (Source: 14b_year_of_crossing.png)

The Eastern Block (C1, C2), by contrast, is better characterized as near-perennial: the shallow till-floored aquifer regularly reaches or approaches the surface in wet winters and flooding is driven by within-season rainfall accumulation rather than antecedent drought depth.

The Forest (C4) prediction equation ($R^2 = 0.865$, summer minimum dominant, winter rainfall non-significant $p = 0.919$; Table 12) is consistent with two non-exclusive interpretations. Under the canopy interception interpretation, summer drought conditions were attenuated by the canopy, producing a more compressed range of summer minima and consequently more predictable winter peaks. Under an alternative interpretation, the forest water table sits sufficiently deep that the winter recharge pulse cannot reach flooding height across the rainfall range in the record, so the regression cannot detect a winter rainfall signal. The two interpretations are not mutually exclusive — both contribute to the statistical pattern — and the available data do not allow them to be partitioned. The P_flood unreachability finding (Section 5.8.1) — that at C4 summer minima of approximately 1.8 m below ground, the required winter rainfall to achieve flooding exceeds anything in the 95-year RAF Valley record — directly supports the second interpretation. The BACI summer minima null

(Section 5.5) is also consistent with a system sitting far below threshold, where neither canopy presence nor canopy removal materially affects whether winter flooding is reached.

The primacy of summer minimum depth is reinforced by the contrasting behaviour of the two ecologically referenced metrics under the climate projection. The Curreli et al. (2013) framework can be applied to two distinct hydrological quantities: the summer minimum, which sets the drought precondition discussed above, and the five-year mean spring water level (MSL5), which describes the sustained baseline within which dune-slack vegetation establishes (Section 4.8.4; van Willegen et al. 2025). The two metrics diverge sharply in their exposure to the projected climate trajectory. Under UKCP18 RCP8.5, the cluster-mean MSL5 shifts by only –21 to –39 mm at the 2080s horizon, whereas the summer minimum deepens by –71 to –134 mm over the same period — a response three to five times larger across every cluster (Section 4.10.1, Figure 69). The mechanism is seasonal: the spring window over which MSL5 is computed receives the projected increase in winter rainfall, which partly offsets the increase in spring evaporative demand, leaving the baseline metric comparatively insulated; the summer minimum, by contrast, receives the full projected increase in summer atmospheric demand compounded by the projected reduction in summer rainfall. The consequence for management is that of the two ecological threshold metrics, the spring baseline is comparatively robust to climate change while the summer minimum is highly exposed. This is why summer minimum depth, rather than the spring baseline, is the binding ecological constraint at Newborough: it is both the precondition for winter flooding recovery and the metric that the projected climate trajectory most directly threatens. The open-dune clusters (C1, C2) that hold a comfortable MSL5 baseline (Section 4.8.4) are nonetheless the clusters whose summer minima the section 4.8.1 trajectory shows approaching the Curreli thresholds — the divergence between the two metrics is not a contradiction but a description of a system whose long-term baseline remains viable while its drought extremes deteriorate.

1.3 Recharge, Loss and the Water Balance

The cluster-level water balance decomposition (Section 3.7.2; Section 4.2.3; Tables 4a, 4b) makes explicit a structural feature that is implicit throughout the state-space model results: under mean climate conditions, the three-term displacement-formulation SSM closes the monthly water balance at the cluster mean level, with residuals of less than 2.5% of total losses (Table 4a). This near-closure indicates that recharge, atmospheric draw and head-dependent drainage together account for the principal water balance fluxes at the cluster aggregate level, with any unmeasured boundary contributions absorbed into the fitted β coefficients (Section 5.3.1) rather than appearing as systematic residuals at the cluster mean. Spatial variation in local closure across the site is examined in Section 5.2.1.

1.3.1 Recharge Partitioning and Atmospheric Draw

Recharge into the water table ($\beta_1 \cdot P$) declines systematically from the Lake Edge cluster (C1: 0.340 m/month) through the Dune (C2: 0.296 m/month) and Western Residual (C3: 0.265 m/month) to the forested clusters (C5: 0.180 m/month; C4: 0.185 m/month), reflecting increasing storage capacity and — in C4 and C5 — the additional suppression of throughfall by canopy interception, which is captured implicitly in the lower β_1 rather than as a separate model term. Total losses (atmospheric draw plus drainage) track recharge closely at every cluster, ranging from 0.186 m/month at C4 to 0.345 m/month at C1. The drainage partition varies markedly: drainage accounts for 85% of total losses at C1 (reflecting rapid lateral exchange with Llyn Rhos-Ddu) but only 24% at C4, where atmospheric draw dominates. C2 and C3 show drainage-dominated partitions (68%/32% and 62%/38% respectively), while C5 is intermediate (60%/40%).

It should be noted that in open-boundary systems, lumped parameter models of this form are known to absorb unmeasured boundary fluxes into the rainfall coefficient when no intercept term is fitted (Young, 2011; Knotters and van Walsum, 1997), which means the recharge term ($\beta_1 \cdot P$) should not be read as a direct measure of local precipitation reaching the water table. At Newborough, however, there is no evidence that such absorption is occurring. The water balance residual field (Section 4.9.7, Figure 58) shows that the three-term balance closes across the site: per-well residual magnitudes are small and uncorrelated with position on either axis, and show no forest-margin or ridge-flank concentration of positive values. The field carries a small systematic negative offset (median -0.0045 m/month; 58 of 66 wells negative). The largest negative residual falls at the open-dune well D7, with the ridge-flank well CEH14 close behind; at both, modelled recharge marginally exceeds modelled losses. Two independent diagnostic tests reach the same conclusion by other routes: a lag-distance test found no relationship between the lag of peak rainfall–residual cross-correlation and distance from the bedrock ridge, and a test for unmodelled evaporative demand using sunshine hours also returned a null result. The caution regarding β_1 therefore stands on general grounds — it remains a lumped coefficient carrying whatever the model does not otherwise represent — but at this site it is a statement about model form rather than a response to any observed residual structure (Section 5.2.1).

Thornthwaite PET is an empirical index of atmospheric evaporative demand derived from temperature and day length; it does not define a specific reference surface in the manner of Penman-Monteith, so a formal crop coefficient decomposition does not apply. The β_2 coefficient under the displacement-formulation SSM therefore absorbs the combined effects of vegetation transpiration, canopy interception, specific yield and depth-coupling into a single effective scaling between atmospheric demand and water table response (Section 5.6.1). C4 and C5 carry broadly similar Corsican pine canopy; the principal difference between them is topographic position and water-table depth, with C4 on the flank of the bedrock ridge (which outcrops within the forest) and C5 on the lower coastal ground. The substantial β_2 difference — C4 considerably higher than C5 — therefore tracks elevation and

the rock base C4 is known to overlie rather than canopy; the depth and extent of that base, and the depth of sand beneath C5, are not constrained without coring. The WTF-derived S_y estimates (Table 4c) show C5's S_y higher than C4's, supporting the direction of this interpretation: a higher S_y at C5 means a given evaporative flux produces a smaller head change at the water table, which would lower the apparent β_2 . The magnitude of the β_2 ratio is larger than the WTF S_y ratio alone would predict if substrate were the only factor, indicating that depth-coupling differences (Section 5.2.2) also contribute. C4's position on the elevated ridge flank is consistent with reduced effective drainable volume per unit head change: the forest overlies the bedrock ridge, which outcrops within the plantation, and an irregular, low-storage rock base — thin sand over a variable bedrock surface — is the natural reading, though its detailed geometry beneath C4 is unconstrained without coring. The storage–drainage index analysis (Figure 49) is consistent with this: C4 wells rank highest in the network on that index, consistent with sluggish lateral discharge rather than free flow through a uniform sand column (Section 5.4.2). β_2 is the atmospheric-draw coefficient, and the high forest β_2 should be read as a property of the ridge-flank setting — elevation, water-table depth and the rock base beneath C4 — rather than as evidence of a boundary input.

A further question the β_2 evidence is sometimes asked to address is whether the Corsican pine canopy provides a shielding effect on the water table: the proposition that the canopy intercepts solar radiation and reduces wind speed at the ground surface, thereby suppressing evapotranspirative demand during summer, when groundwater is already at its seasonal minimum and ecological stress is highest.

One coefficient-level observation is consistent with the proposition. C5 Coastal Forest's β_2 (1.31) lies below the open-dune range (C2: 1.76; C3: 1.83): per unit of Thornthwaite PET, the water table beneath the Coastal Forest experiences less drawdown than equivalent open-dune ground. C4 gives no comparable support. Its β_2 (2.58) stands well above the open-dune band, and the identifiability diagnostic (Section 5.2.3) establishes that this is a real property of the cluster rather than an artefact of a poorly resolved drainage term. Since the two clusters carry broadly the same canopy, the two-fold β_2 contrast between them cannot itself be a canopy signal. The C5 comparison is also not strictly like-for-like: because the SSM is fitted on gross rainfall and above-canopy PET, β_2 at the forested clusters carries the canopy's interception loss as well as sub-canopy demand (Section 3.7.2), so C5's β_2 sits below the open dune despite carrying a rainfall tax that C2 and C3 do not.

The annual β_2 coefficient is in any case a weak instrument for the question, and two diagnostics already run bear this out. The depth-dependent ET formulation (Section 3.7.1) returns the network's weakest depth coupling at C4 ($\lambda = 0.15 \text{ m}^{-1}$) alongside its highest β_2 , so the elevated coefficient is not a depth-mediated evaporative effect. The residual seasonality diagnostic finds no unmodelled summer evaporation anywhere on the site: the correlation between SSM residuals and independent sunshine-hour data is negligible across all clusters (network mean $r = -0.026$; no well outside the Bartlett null band), and no well in

the network carries a summer-phased residual peak, 48 of 63 peaking instead in winter or early spring. Whatever systematic bias the Thornthwaite formulation carries relative to actual forest evapotranspiration, β_2 has already absorbed it.

There is also a reason to expect any real pine–dune contrast in evaporative loss to be winter-weighted rather than summer-weighted, and it is a matter of rainfall volume rather than plant phenology. The canopy's principal hydrological cost is interception: approximately 24% of incident rainfall is caught and re-evaporated before it can reach the water table (Freeman, 2008). That loss scales with rainfall, and rainfall at Newborough is strongly winter-dominated, so the difference in water reaching the ground between forest and open dune is at its largest in winter. In summer, when rainfall is much lower, the interception difference between the two — and the evaporative loss that follows from it — is correspondingly smaller. A rainfall-scaled, winter-weighted loss represented through a coefficient on PET, a summer-weighted quantity, is represented awkwardly at best. The annual β_2 contrast between forest and open dune is therefore not a measurement of summer canopy shielding, and should not be read as one in either direction.

One confound attaches to the C5 evidence. The Coastal Forest cluster carries a long-term coastal-retreat gradient (Section 4.8.2; Section 5.7.2) that is embedded in the fitted coefficients: the lower C5 β_2 may partly reflect a secular convergence of the water table toward a lower baseline driven by shoreline retreat rather than a canopy effect on evaporative demand. Separating these mechanisms in C5 is not possible without a longer record or unforested coastal-margin control wells alongside the forested ones. The shielding interpretation of C5's β_2 is therefore consistent with but not uniquely supported by the data; the evidence for canopy effects on atmospheric draw is suggestive rather than conclusive.

The clearfell record bears on the proposition from a different direction and is discussed in Section 5.5.1. Its seasonal asymmetry — a significant annual rise following felling that is not distinguishable from zero over the summer months — runs in the direction a real shielding effect would predict, but the summer estimate is too imprecise to establish it. Taken together with the C4–C5 contrast under common canopy and the interception confound above, the shielding proposition remains open on the present evidence rather than supported or refuted by it.

1.3.2 Drainage Decay and the Volumetric Water Balance

The drainage decay coefficient β_3 controls the rate at which the water table relaxes toward its equilibrium position, and its spatial distribution (Section 4.9.2, Figure 48d) shows the strongest contrast of any SSM parameter across the site. C1 Lake Edge drains fastest, driven by rapid lateral exchange with Llyn Rhos-Ddu. C2 and C3 show similar intermediate rates. C5 Coastal Forest is lower, and C4 Main Forest has the weakest drainage in the network (Table 3). The displacement formulation (Section 3.4.1) is essential to resolving the C4 drainage term: under a ground-surface reference C4's β_3 was borderline significant;

referencing drainage to the 3.7 m datum resolves it to a clearly significant value (Table 3). The slow drainage at C4 is physically consistent with the canopy suppressing recharge pulses that would otherwise drive the water table above its equilibrium and generate a strong drainage response, combined with sluggish lateral discharge through C4's inferred irregular, low-storage base (Section 5.3.1; Section 5.6.1). The contrast between fast-draining eastern clusters and slow-draining forest clusters shapes the volumetric water balance partition between drainage and atmospheric draw.

The indicative volumetric water balance (Table 4b) provides a complementary perspective. All clusters receive approximately 892 mm/yr of rainfall; for the forested clusters, canopy interception (24%; Freeman, 2008) reduces the throughfall reaching the ground surface to approximately 678 mm/yr. The ET/drainage partition is bracketed by two independent methods — the SSM β_2/β_3 ratio and a seasonal recession curve analysis — with the mid-point reported in Table 4b. The cluster partitions differ substantially: at C1, drainage dominates total losses, reflecting rapid lateral exchange with the lake; at C4, atmospheric draw dominates losses and drainage is weak, indicating that the Forest cluster's water balance is governed primarily by the competition between rainfall input and evaporative demand rather than by hydraulic drainage. C2 and C3 sit intermediate, with drainage-dominated partitions; C5 is closer to balanced.

The water table fluctuation (WTF) analysis (Section 4.2.4; Table 4c; Figure 12) addresses the question of whether the substrate's storage capacity differs across the cluster structure. Uncorrected forested cluster S_y medians exceed the open dune range, but this reflects the WTF method attributing the full rainfall total to recharge when in reality a fraction is intercepted by the canopy. Once corrected for canopy interception, C4's S_y estimate falls within the open dune range and C5's sits at its upper edge, though C5's is constrained by the plausibility clip and is only weakly corroborative. The effective storage inferred from the water-table response is therefore broadly uniform across the site — consistent with, though on these data not proof of, a well-mixed aeolian depositional history — so the forest clusters' distinctive β -coefficient signature (lower β_1 , higher LCSC) is read as deriving largely from canopy boundary conditions at the surface rather than from differences in subsurface storage architecture, though depth-coupling and lateral drainage geometry also contribute to within-forest β variation (Section 5.3.1).

The implication for management interpretation is that the surface-mediated canopy effects identified by the analysis are in principle responsive to silvicultural intervention, while the subsurface storage and transmission properties of the aquifer are fixed by geological history. The net hydrological response to silvicultural intervention at the cluster level is small relative to climate forcing over the same horizon (Sections 5.5.2, 5.7.1), and the spatial reach of forest management does not extend to the open dune clusters (Section 5.8); the surface-vs-subsurface distinction therefore concerns *which* of the canopy's effects are accessible to management rather than *whether* canopy management is the dominant restoration lever.

The slow drainage and thin-aquifer substrate identified above also explain a finding that bears directly on the shielding hypothesis: the inter-annual amplification field (Figure 65; Section 4.9.8) shows C4 Main Forest to be the most climate-sensitive zone on the site, with a mean spring water-table swing approximately 1.72 times the network mean between the canonical dry and wet year sets (range 1.38–2.35×). The forest interior also occupies the hydraulic high of the aquifer, furthest from any lateral discharge boundary — the lake to the east, the coast to the south-west, the estuary to the south-east — so the water table there rises and falls without the damping that proximity to a constant-head boundary provides to the lake-edge and coastal clusters. The lake edge damps the shared climate swing to 0.61× — the lake providing a hydraulic backstop in dry years and an overflow in wet ones — while the open dune sits near unity. The amplification coefficients correlate negatively with β_3 ($r = -0.45$, $p < 0.001$): slower-draining wells retain wet-year recharge longer and therefore show larger spring swings between dry and wet years. This is the same mechanism identified in Section 5.3.2 above — low S_y over an irregular bedrock base produces a thin saturated column that amplifies any given volume of recharge or discharge into a proportionally larger head change.

This amplification result runs counter to the shielding hypothesis as a statement about dynamic climate buffering. A canopy that meaningfully shields the ground surface from climatic forcing would be expected to damp inter-annual variability, not amplify it. The forest interior over-responds to climate swings relative to the open dune. Two positional factors account for this without invoking canopy effects, and both are consistent with the WTF S_y analysis and the slow-drainage evidence above: the thin saturated column over irregular bedrock (low effective S_y amplifying head change per unit flux), and the forest's position at the hydraulic high of the aquifer, remote from any constant-head boundary that would buffer wet-year rises or dry-year falls. The forest canopy and the bedrock substrate are spatially collinear at Newborough Warren — the plantation occupies the ridge-proximal zone where the aquifer is inherently thin — and their contributions to the amplification signal cannot be separated from the dipwell record alone. Independent substrate characterization beneath C4 (slug tests; cored transect) and a targeted seasonal β_2 decomposition comparing summer and winter months across pine and open-dune wells are identified as the analyses most likely to resolve the question in future work.

The amplification result does not, however, constitute evidence that the canopy provides no hydrological buffering and can therefore be removed without hydrological cost. The amplification field (Figure 65) measures absolute inter-annual swing — the difference between wet-year and dry-year spring water levels at each well, expressed as a ratio to the network mean. It is a structural property of the aquifer at each location: a well sitting at the hydraulic high over thin, slowly draining sand will amplify any given climatic forcing regardless of what vegetation covers it. The relevant question for buffering is not whether the forest zone swings more than the open dune in absolute terms — it does, for structural reasons — but whether the canopy modulates the response relative to what the same aquifer

location would do without it. That question is answered on the relative (BACI) scale, not the absolute one. The BACI climate-sensitivity analysis (Figure 28; Section 4.6.2) operates on exactly this scale: it compares the felled well against matched controls sharing the same climate forcing, so the structural amplification signal — common to felled and unfelled wells alike — cancels. The significant $CWB^2 \times \text{clearfell}$ interaction (Impact: -27.5 mm per $(100 \text{ mm CWB})^2$, $p = 0.016$; Edge: -22.2 mm per $(100 \text{ mm CWB})^2$, $p = 0.008$) indicates that after canopy removal the felled wells responded more strongly to dry-period deficits than before, relative to their controls. Under the canopy, the felling-attributable rise was largest in dry conditions — the pattern a canopy-buffering mechanism operating specifically at the dry tail would produce. This curvature result is reported as a preliminary finding (Section 4.6.2) and rests on a single felling event, but it provides the only evidence in this study that operates on the BACI relative scale and is therefore not confounded by the structural amplification identified above. The amplification field and the BACI curvature result therefore address different questions: the former characterizes where the aquifer is structurally sensitive to climate; the latter provides the most direct available test of whether the canopy modulates that sensitivity. They are not in conflict, and the amplification field should not be read as evidence against canopy buffering.

In summary, the cleaner and less confounded evidence for a canopy effect on the dune-slack water balance remains the recharge suppression captured in the lower β_1 of both forested clusters — a consistent finding across the WTF, LCSC and SSM analyses that is not materially affected by substrate or coastal-retreat confounds. The summer-shielding component, assessed through β_2 , is suggested by C5's coefficient and the corrected C4 value, but the amplification result and the substrate/coastal confounds mean it cannot be stated as a confirmed finding on the present evidence.

1.4 Dune Scraping

Three dipwells — CEH36, CEH18 and CEH21 — were subject to ground surface lowering by mechanical scraping, with CEH36 scraped in April 2015 and all three re-scraped in October 2023 (Section 4.5). CEH36 is located approximately 340 m seaward and downslope of the December 2017 clearfell footprint, a spatial relationship that becomes important when the interaction between the two interventions is considered below.

1.4.1 Measured Benefit and Durability

At CEH36, three independent methods (raw BACI, synthetic control, SSM forward residual) all identify a scraping benefit in the same direction during the pure scraping era from April 2015 to December 2017, with the raw-BACI and synthetic-control estimates in close agreement and the SSM forward-residual estimate lower but of the same sign (Section 4.5; Figures 18, 19). The seasonal record reinforces the finding: the era-mean summer minimum shallowed substantially during the scraping era and held its position through the post-felling

period, even as the CEH4 control deteriorated under the background drying trend (Section 4.5; Figure 21). Calibrated against the Curreli et al. (2013) eco-hydrological gradient (Section 5.2), the summer improvement at CEH36 represents a shift across more than half the gradient separating the SD15b and SD16 community thresholds.

Scraping creates a topographic low that acts as a local drain, raising the water table at the scraped site by collecting lateral inflow. The whole-array step map for the scraping era shows no coherent uphill gradient in the observed level change: across the wells uphill of CEH36 the climate-corrected steps scatter in both signs, without the distance-decay a signal radiating from a point source would produce (Section 4.5; Figure 25). At WMC3 specifically, however — the one uphill well with a full pre-scraping record and a clean forest-control comparison — a level drawdown is resolvable and reproducible. Its climate-corrected step during the pure-scraping era is -46 mm (Figure 25), and its displacement gap against the forest control falls by 55 mm at the April 2015 scraping and by a near-identical 54 mm at the October 2023 re-scraping (Section 4.6.3, Figure 29) — the consistency across two independent events, from two different control choices, being the signature a genuine drainage effect would produce. This level signal is distinct from the drainage-coefficient response: the separate $\Delta\beta_3$ propagation test found no distance-decay in the drainage coefficient (Section 4.5.5), and the two results are compatible, since a head drawn down toward the lowered scrape surface need not register as a change in a well's intrinsic drainage rate. The near-field drainage half-life — 13 to 21 months at wells within ~ 260 m of CEH36 — is short enough for such a level signal to appear within the record, in contrast to the far-field equilibrium magnitudes considered in Section 5.4.3. The benefit at the scraped site does not alter the rate at which the water table responds to climate forcing, so the background drying trend continues to erode the initial improvement; at CEH36, the improvement is diminishing but the gain has been retained through the post-felling period to date. CEH18 and CEH21, scraped only in October 2023, have shown limited and inconsistent responses that do not survive Tier 1 correction for background drift — consistent with their position closer to the coast where progressive erosion outpaces the localized topographic benefit (Section 4.5).

The shape of this diminishing-but-retained trajectory is itself informative. Against the climate-corrected control the summer-minimum gap at CEH36 rose to a peak of about +422 mm roughly two years after the April 2015 scraping, then relaxed — at approximately -17 mm yr^{-1} measured from the peak, or -14 mm yr^{-1} from the scraping year — toward a residual of about +187 mm, a partial decline that leaves the slack well above its pre-scrape state (Section 4.5). Three mechanisms acting on different timescales can account for this shape. Removing the dune-slack vegetation removes a growing-season transpiration draw, which would tend to raise the local table; this competes, to a degree not yet resolved, with the newly exposed bare sand and standing water at the scrape floor adding direct evaporation, which would tend to lower it — the net sign of this mechanism is not established from the present evidence. This is not the same physics as the clearfell canopy-

removal step: the clearfell's benefit is attributed to rainfall reaching the ground once canopy interception is removed (Section 5.6), a mechanism with no counterpart at an unforested slack, and one that registers only in the monthly mean — the clearfell's own summer-minimum step is close to zero (Section 4.6.4) — so the clearfell cannot serve as a validating precedent for a summer-minimum-specific effect at the scrape. Cutting headward into the landward flank creates a new low point in ground where the inland head is higher, which steepens the gradient into the scrape floor: inflow is high initially and then relaxes as the enhanced inflow draws the surrounding head down and the gradient flattens — the transient expected on relief-drainage grounds (Hooghoudt, 1940; van Schilfgaarde, 1963; Freeze and Cherry, 1979) and quantified for a comparable fine-sand dune–swale setting as largely completing within about five years (Gerla, 2019). As vegetation recolonises the exposed floor, transpiration returns and the balance of the two competing surface losses shifts back toward the vegetated state — eroding whatever net benefit the first mechanism had contributed, while also reducing the offsetting bare-surface evaporation term as the floor is shaded and stabilised again. The observed curve is consistent with an initial overshoot that relaxes and is then slowly modified by regrowth; the residual reflects this recolonisation alongside the drainage relaxation of the second mechanism, and cannot be decomposed into the two contributions with the present data. The clearfell offers a useful contrast in shape, though not in mechanism: it involves no excavation, and in the matched summer-minimum currency the felled-well response shows no comparable relaxation (a post-felling trend of about $+1 \text{ mm yr}^{-1}$, indistinguishable from zero; Section 4.6.4). Because the clearfell's own summer-minimum step is itself close to zero, this flat trajectory shows only that the observed decay at CEH36 is specific to the scrape rather than a generic post-intervention artefact — it does not isolate a transpiration-only, decay-free vegetation mechanism, since the clearfell's measured effect operates through a different pathway (interception recovery) and a different metric (the monthly mean). This excess decay is consistent with the recharge-capture transient, but it cannot be attributed to relaxation alone: the decline is present against the climate control yet absent against the CEH4-paired control (about $+7 \text{ mm yr}^{-1}$, not significant), because CEH4 lies at the seaward end of the slack and is subject to the same coastal drawdown, so pairing cancels the shared coastal signal. The relaxation is therefore reported as consistent with, not diagnostic of, the drainage-theory mechanism, given the single scraped well and this currency dependence. Unlike the clearfell, whose recovery is confined to the monthly mean and does not reach the summer minimum or the five-year spring baseline (Section 4.5.4), the scraping benefit at CEH36 is expressed across all three metrics. This broader pattern is more consistent with a mechanism operating through the growing season — transpiration removal and/or the geometric drainage effect described above — than with a rainfall-triggered interception-recovery pathway of the clearfell's kind, though it does not on its own resolve which of the two dominates. These inflow statements concern the scrape floor, which the inflow wets — an on-site effect, distinct from the surrounding drawdown considered in Section 5.4.3.

1.4.2 Interaction with the Clearfell

The interaction between scraping and the subsequent clearfell is instructive. CEH36 sits approximately 340 m downslope of the felling footprint, connected by the continuous downslope drainage pathway identified in the Darcy flow analysis (Section 4.9.6, Figure 57). The CEH36 BACI step during the felling pulse era was smaller than during the pure scraping era — the scraping benefit diminished but was not reversed (Section 4.5; Figure 24). The CEH4 control deteriorated across the same period under the background trajectory, and a substantial part of the apparent reduction in BACI step at CEH36 reflects CEH4's own deterioration rather than a loss at the scraped well. Any recharge benefit from the clearfell propagating downslope to CEH36 would operate on the multi-year to multi-decadal timescale developed in Section 5.4.3 and may not yet be detectable in the monitoring record.

The reverse interaction also warrants consideration: the April 2015 scraping at CEH36 created a topographic drain that may have drawn water away from the wells used to measure the clearfell response. WMC3 (the sole Impact well) and CEH31 (the nearest Edge well) sit closer to CEH36 than the other Edge wells, and the scraping drain at these positions is directly observed, not merely inferred: at WMC3, 262 m from CEH36, the raw displacement gap against the forest control falls 55 mm at the 2015 scraping and 54 mm at the 2023 re-scraping (Section 4.6.3), the near-identical steps across two independent events being the signature a genuine drainage effect would produce. This provides observational support for treating the fitted clearfell step as a residual net of the scraping drain as well as CEH4's own deterioration. The combined footprint of the scrape drain and the coastal-retreat gradient on the pre-fell baseline is mapped in Figure 56; it is the spatial counterpart of the distance-weighted scraping covariate and the easting $\times$ time term (Section 4.6), making explicit that the fitted clearfell step is a residual net of two non-felling drawdown processes. The two interventions propagate through the same continuous aquifer on overlapping timescales, and the interaction illustrates the difficulty of isolating individual management signals in a system where perturbations are entangled.

Scraping at CEH36 produced a benefit in summer minimum depth that persisted through the post-felling period. The benefit diminished but was not reversed; the dominant threats to scraping durability identified by the monitoring record are the background climate drying trend and coastal erosion at the western boundary, rather than upstream canopy management.

1.4.3 Coastal Erosion Analogy and Cascade Risk

The mechanism by which scraping produces its measured benefit — lowering the ground surface to reduce the depth to water table — is physically identical to the mechanism of coastal erosion. Both reduce ground elevation relative to a fixed absolute water table position; both increase the displacement head (h_{disp}) at adjacent wells; and both consequently enhance drainage gradients away from the intervention or erosion zone. The Tier 1 CUSUM trajectories at CEH4 and CEH22 (Section 4.5; Figure 17) show this process

operating naturally at the western coastline, with both control wells showing accelerating negative CUSUMs attributable to progressive coastal erosion of the dune edge. That natural process is now resolved spatially. The coastal-margin field of Section 4.9.4 (Figure 54) maps a narrow erosion-dominated band of net water-table deepening along the dune-toe margin, coincident with the western C5 and C3 wells; its driving shoreline retreat is documented independently by Pye and Blott (2024), whose dune-toe sequence for the Twyni Penrhos frontage records up to 16.3 m of retreat at the southern end between 2013 and 2022 — of the order of 1.8 m yr^{-1} at the most active location — placing a measured floor under the rate the field assumes.

This analogy carries a management implication that the SSM physics make explicit but the available monitoring record can only partly test. Scraping along the western coastline, while locally beneficial in reducing the depth to water table at the scraped location, increases the hydraulic gradient between the coast and the dune interior. The SSM drainage term operates in both directions: a scraped coastal zone with elevated h_{disp} draws water seaward, deepening the water table at wells uphill of the scrape. This dipole — a rise at the scraped slack, drawdown of the surrounding water table — is given spatial form in Figure 55, which renders the scrape as a steady-state drain anchored to the measured CEH36 response. The near-field level drawdown is resolvable in the record: the WMC3 displacement-gap steps of Section 4.6.3 register it directly, and the $\lambda \approx 230 \text{ m}$ decay model places the predicted equilibrium head at 262 m at -42 mm , of the same order as the observed 55 mm step (Section 5.4.1). The off-site magnitudes below, expressed as a volumetric flux (mm water-equivalent per month), remain modelled steady-state values the record cannot resolve at these distances. The scenario framework puts an indicative magnitude on this neighbour drawdown: modelled from the same drain cone (Figures 26–27; Section 4.5.6), the off-site drawdown at a well 100 m from a CEH36-equivalent scrape is of order -30 mm water-equivalent per month, over the $\lambda \approx 230 \text{ m}$ length scale. These are steady-state figures on the same footing as the uphill drawdown of Figure 55, not values the 31-month record can confirm, but they indicate the neighbour effect is a modest fraction of the on-site benefit rather than negligible. The magnitude of this effect is governed by the same aquifer parameters that control the natural coastal erosion signal — and the results at CEH4 and CEH22 show that signal to be substantial and progressive.

The risk of a cascade follows from the mechanism. If coastal scraping deepens summer minima at interior wells, those wells may in turn be identified as candidates for scraping — lowering their ground surface, increasing their h_{disp} , and propagating the drainage enhancement further inland. Each successive scrape addresses the local symptom while intensifying the systemic cause. This chase-inland dynamic is the topographic analogue of the drainage acceleration that the scraping analysis identified as a fitting artefact in Section 4.5.3: the aquifer is not gaining storage decay capacity, but the systematic lowering of ground surfaces produces a head trajectory that the model — and the manager — interprets as progressive deepening requiring further intervention. The BACI-corrected $\Delta\beta_3$ values at

uphill wells (Section 4.5; Script 09b) showed no distance-decay from the scraping site within the observation window. This null concerns the drainage *coefficient*, and is consistent with a pre-existing spatial gradient rather than a propagating β_3 signal (Section 4.5.5); it is not evidence that the cascade mechanism is inactive. The *level* drawdown, by contrast, is resolvable in the near field — the WMC3 displacement-gap steps of Section 4.6.3 register it at 262 m — while the far-field equilibrium magnitudes remain modelled figures the record cannot confirm.

Most of the Western Residual already lies below the hydrological range for functioning dune slack habitat (Section 4.8; Figure 45), so the practical scope for this cascade is limited to the transitional margins. The principle that coastal scraping mimics coastal erosion in its effect on the inland water balance is well-founded in the SSM physics regardless of the available data's ability to test it within the observation window, and on the precautionary principle should inform the spatial planning of future scraping programmes. The most productive scraping targets identified by the analysis (Section 5.7) are the C1/C2/C3 transitional wells in the mid-eastern sector, where structural sensitivity maximizes the retained benefit and where the scraping does not amplify an existing coastal drainage gradient. This recommendation is reinforced by the spatial asymmetry of the shoreline process itself. Pye and Blott (2024) document a north-positive (progradation) to south-negative (retreat) gradient along the frontage, with the chronic retreat concentrated at the southern end and the erosion front migrating northwards. Coastal scraping sited at the actively retreating southern margin would coincide with a location already undergoing net retreat, and would there behave less like a controlled, durable surface-lowering than like an addition to that retreat — a further argument for confining scraping to the mid-eastern transitional wells away from the eroding south-western margin.

The geometry of the scrapes reinforces this precaution. The cuts are elongated fingers projecting inland from the eroding coast, aligned with the coast-to-inland hydraulic gradient and discharging seaward at the lowest available base level; whatever their habitat intent, this is an effective relief-drainage configuration, disposed to lower the water table in the dune around each finger even while the finger floor itself is wetted by inflow (Section 5.4.1). The retreating coast does not lengthen the fingers — erosion shortens them from their seaward end — but it lowers the base level to which they drain, so the surrounding drawdown a finger imposes and the coastal-erosion drawdown act in the same direction and cannot be separated within the record. This drainage-engineering dimension of scrape design appears to be an unexamined one. Conservation guidance recognises that creating scrapes or pools can dry adjacent dune slacks (Buglife, 2019) but frames this as a siting caution without a mechanism, and otherwise treats scraping as cut-to-the-water-table habitat creation (Grootjans et al., 2002; Jones, Rooney and Rhymes, 2021); we are not aware of guidance that applies relief-drainage theory to scrape design, or that relates a scrape's geometry — its length, its orientation to the hydraulic gradient, and its discharge to a falling seaward base level — to its severity as a relief drain on the surrounding dune. The mechanism set out here

offers a candidate explanation for that recognised but unexplained risk, and it reinforces the case for siting scrapes among the mid-eastern transitional wells rather than along the retreating south-western margin.

The timescale over which these processes operate is governed by the aquifer's hydraulic diffusivity. Using representative values for clean dune sand and the interception-corrected WTF Sy ranges from Section 4.2.4, the characteristic propagation timescale at 340 m is years to a decade, and at 570 m is one to several decades; the literature values for unconfined dune sand (Fetter, 2001) shorten these ranges but do not alter the order of magnitude. A direct measurement of the inland reach of the coastal-erosion gradient from the 21-year monitoring record (Section 4.8.1; Figure 41) gives a fitted strip-aquifer width of 894 ± 51 m under a Dupuit–Forchheimer steady-state interpretation, or an equivalent exponential decay length of 413 ± 65 m under a transient diffusive interpretation. Both fall within the diffusivity-derived range above, providing independent network-scale corroboration of the years-to-decades timescale regime; and the modelled coast-edge rate is itself independently corroborated by a model-free coast-to-inland transect (Section 4.8.3), whose -28.2 mm yr^{-1} growth in the coast-minus-inland head difference matches the -29.0 mm yr^{-1} gradient δ_0 from an entirely separate, collinearity-free construction. Whichever estimate is adopted, the propagation timescale is measured in years to decades. This has three consequences. First, it explains why the BACI-corrected $\Delta\beta_3$ at uphill wells showed no distance-decay: the propagation timescale exceeds the 31-month observation window. Second, the coastal erosion signal visible in the CEH4 and CEH22 CUSUMs is a lagged response to erosion that occurred years earlier; if the rate of coastal retreat is itself accelerating, consistent with regional projections under sea level rise (Pye and Saye, 2005) and the CEH22 trajectory from 2021, then the full hydrological consequences of recent erosion have not yet propagated to the interior wells. The current CUSUM trajectories are a leading indicator rather than a real-time measure of present inland conditions. Third, it provides a physical explanation for the clearfell's slow equilibration, consistent with the propagation timescale combined with C4's weak drainage. The aquifer's response to any boundary perturbation — whether erosion, scraping or land cover change — is measured in years to decades, and monitoring programmes need to be designed accordingly.

Sea-level rise is the opposing process acting at the same western boundary (Figure 53). The Holyhead tide gauge records a historic mean-sea-level rise of 2.3 mm yr^{-1} (Pye and Blott, 2024), with recent satellite altimetry suggesting in excess of 3 mm yr^{-1} in the north-east Atlantic. The western C5 and C3 wells therefore lie under two coastal pressures of opposing sign — chronic dune-toe retreat and gradual sea-level rise — whose net effect Section 4.9.4 shows to be erosion-dominated along the dune-toe margin, reinforcing the monitoring priority for the western-margin wells.

The direction of this coastal influence — shoreline retreat narrowing the dune body and lowering the water table — is consistent with the behaviour documented at comparable

Welsh and north-west European dune systems, though the present network cannot independently confirm its amplitude here (Section 4.9.8). The closest analogue is Whiteford Burrows, a natural dune-slack spit on the Gower coast of South Wales, where Robins, Pye and Wallace (2013) found that erosion narrowing the sand body lowered the dune water table by approximately one metre following a loss of only about 4% of the spit width — a mechanism and direction precedent for the erosion-drawdown process inferred here, not a rate comparison. Two further systems bracket the behaviour. The engineered Sand Engine nourishment in the Netherlands (Huizer et al., 2018) shows the same coupling between sand-body geometry and freshwater-lens elevation in the opposite, net-growth direction, and the natural barrier island of Dog Island, Florida (Kiflai and Whitman, 2023) demonstrates that the water-table response to shoreline change can lag the erosion itself by years to decades — a disequilibrium that is one reason the modelled coast-to-inland profile of Section 4.9.4 need not track the present-day retreat rate exactly. This erosion-drawdown mechanism is the inverse of the more commonly cited expectation that sea-level rise raises coastal water tables (Ferguson and Gleeson, 2012; Michael et al., 2013): at Newborough the dominant coastal signal at the western margin is a lowering driven by dune-toe retreat, not a rise driven by the sea, and the two act in opposing directions as Section 4.9.4 sets out. Stating this inversion explicitly is more faithful to the evidence than presenting the sea-level-rise expectation as the default.

1.5 The Clearfell Experiment

The December 2017 clearfell of approximately 8.4 ha of Corsican pine plantation provides the most direct test of how canopy removal affects the water table in the dune slack system. The clearfell zone sits upslope of CEH36 (the scraped slack described in Section 5.4) and straddles the gradational boundary between C4 Main Forest and C5 Coastal Forest. The sole Impact well, WMC3, shows strongest Pearson affinity with C3 Western Residual ($r = 0.965$) with C5 a close second, and the extended-network FE wells within the felled footprint show similarly mixed affinities across C3, C4 and C5 (Section 4.3). The felled compartment therefore cannot be treated as a clean C4-only experiment; the clearfell zone spans the transition from forest to open dune, and its hydrological effects propagate downslope into the C5 catchment via the continuous drainage pathway identified in the Darcy flow analysis (Section 4.9.6, Figure 57).

The three-counterfactual ANCOVA-BACI (Section 4.6.2) returned a significant clearfell step against the Forest control — the most direct test, comparing felled wells against unfelled forest sharing the same canopy and substrate — of +120 mm at the Impact well (WMC3; $p < 0.001$). At the Edge tier the step was +33 mm and not statistically distinguishable from zero ($p = 0.19$). The measurable clearfell recovery is therefore localized at the felled well rather than expressed across the surrounding edge of the compartment; an alternative reading — that the edge wells are undergoing a continuous decline that a step-

change BACI is not well suited to detect — cannot be excluded on the present record, and is taken up in Section 5.5.1. The Climate-control comparison captures the standing forest's pre-existing hydrological footprint alongside the December 2017 clearfell; the Forest control is the appropriate test for canopy-removal-specific attribution.

Four independent robustness methods support the ANCOVA result: SSM residual normalization, synthetic control via donor weighting, synthetic extension of FE1 and FE2 within the felled compartment, and a clearfell transect regression (Section 4.6.7). All four return clearfell steps in the same direction as the ANCOVA and at comparable magnitudes. The robustness estimates sit below the ANCOVA's Impact-well step because they do not include the easting $\times$ time correction for the coastal erosion gradient, and so represent lower bounds on the clearfell-specific component. The transect regression returned no significant spatial gradient, although the post-felling spread in transect hydrographs is visibly wider than pre-felling, with WMC3 consistently above the pack — suggesting a distance-dependent response that the transect regression lacks the power to resolve. The summer minima analysis (Section 4.6.4) returned no significant clearfell effect at the Impact tier and a small negative effect at the Edge tier, opposite in sign to the monthly ANCOVA result — summer minima at the felled Edge wells deepened slightly relative to the Forest control after felling. The clearfell therefore registers in the mean monthly water level but not in the summer extremes, where felled and unfelled wells track each other. The seasonal partition reflects an asymmetric outcome of the canopy's recharge-suppression and atmospheric-shielding effects. Canopy removal raises β_1 and delivers a substantial winter recharge gain at the felled wells, while the corresponding rise in β_2 would impose a summer drawdown cost but is largely offset by the site-wide β_1 decline operating on felled and unfelled wells alike. The +120 mm mean lift therefore arises predominantly from the winter recharge recovery, with summer minima essentially unchanged. The β_1 and β_2 contributions to this partition are quantified in Section 5.5.1, where the role of the site-wide β_1 decline operating disproportionately on summer lows is developed.

Independent validation of the ANCOVA's easting $\times$ time correction against the network-scale coastal-retreat gradient model (Section 4.6.3, Table 8) shows the Forest Impact comparison sits within sampling error of the model prediction ($z = -1.16$), confirming that the +120 mm Impact-well clearfell step is not an artefact of incomplete gradient correction. The Forest Edge comparison is likewise consistent with the gradient model ($z = -0.29$). The covariate therefore absorbs a quantity of west-to-east deepening at both Forest-control comparisons that matches the independently estimated coastal-retreat signal, rather than over- or under-correcting it.

The coefficient decomposition (Section 4.6.6; Table 11; Figure 35) provides additional context. Recharge sensitivity (β_1) declined across all five tiers of the BACI network — a site-wide phenomenon visible in controls and treatment wells alike, ruling out a clearfell-specific origin (mechanism discussed in Section 5.5.1). Atmospheric draw (β_2) increased at

Forest Control wells, consistent with progressive canopy maturation, but showed mixed responses at Climate Controls. Storage decay (β_3) changes were small and spatially incoherent. The predicted clearfell effect from coefficient shifts alone was negative at the felled tiers, but because controls show comparable shifts, the net coefficient-attributable component is near zero. This sits alongside the ANCOVA's significant +120 mm Impact-well step. The two analyses ask different questions. The ANCOVA measures the gap between felled and control wells, so the site-wide β_1 decline — which affects both — cancels and the canopy-removal effect is visible. The per-well decomposition measures coefficient changes at each well individually; the β_1 decline registers as a negative contribution at every well, and because controls and felled wells show comparable shifts, no clearfell-specific signal emerges. The two are reconciled rather than contradictory.

1.5.1 Seasonal Asymmetry, the β_1 Decline, and the Confounded Baseline

The site-wide β_1 decline is a significant finding independent of the clearfell result. It represents the strongest candidate mechanism for the observed summer minimum decline documented in Section 4.8.1 and is independent of any management intervention. The decline is most parsimoniously attributed to changing within-month rainfall distribution. If the same monthly total arrives in fewer, more intense events separated by longer dry spells, water held in the unsaturated zone between events is increasingly drawn back out by ET before it can percolate to the water table. The net recharge per millimetre of rainfall therefore falls even where the gross monthly total is unchanged. The mechanism is measurable at this site. Freeman (2008) recorded the open-dune profile at roughly 17.7% volumetric moisture against 6.5% beneath the plantation, with the open-dune upper 20 cm held nearly static by capillary contact with the water table. Where that contact is maintained, a lengthening inter-event interval draws the profile down before the next pulse arrives, and the recharge efficiency of a given monthly total falls accordingly. The evidence is one season at three plots and is offered as a supporting analogy rather than a test of the site-wide decline.

The magnitude of this site-wide decline is substantial. Across the seventeen wells of the BACI network (Section 4.6.6; Table 11), the mean β_1 recharge coefficient fell from 2.81 in the pre-felling era to 2.65 in the post-felling era, a network-mean reduction of 5.8%, with fourteen of the seventeen wells showing a decline. The decline is observed at the unfelled Climate Control tier (mean -3.8%, four of five wells), sited at distance from both the clearfell footprint and the coastal margin, and at the unfelled Forest Control tier (mean -2.0%, three of five wells); the larger Edge-tier and Coastal Control declines (-10.2% and -13.1%) absorb indirect contributions from post-fell recovery dynamics in the immediate clearfell vicinity and from the coastal-retreat gradient documented in Section 4.8.2 respectively. The Climate Control tier provides the cleanest evidence that the β_1 decline operates independently of canopy removal and is a genuinely site-wide trend.

Against this background, the clearfell produces an inherent seasonal asymmetry. Removing the canopy recovers the interception loss, raising β_1 and increasing winter recharge. The same removal increases β_2 — the water table's sensitivity to atmospheric demand — so a given monthly PET produces a larger summer drawdown. The two effects operate on different seasons and through different pathways; their net annual balance is the quantity of management interest, and is addressed in Section 5.5.2.

The CWB $\times$ clearfell interaction — testing whether the water table's sensitivity to cumulative climate forcing changed after felling — was non-significant in all six ANCOVA models (Section 4.6.2). The climate sensitivity of the felled wells did not detectably change after canopy removal: the ANCOVA detects a significant upward shift in mean water table level, while the rate at which the BACI gap responds to cumulative water balance deficit remains unchanged, with pre-felling and post-felling slopes approximately parallel (Figure 28). This is consistent with the canopy acting simultaneously as a hydrological sink (transpiration, interception) and a microclimatic buffer (shading, reduced wind exposure, organic forest floor). Canopy removal modified both functions: the recharge gain raised the mean water table, and the concurrent change in atmospheric draw operated on the same monthly water balance, so that the system's net sensitivity to cumulative climate forcing was preserved. The practical consequence is that the clearfell effect on the BACI gap is a fixed offset — it shifts the baseline from which the water table responds to climate — rather than a change in the rate of response to wet or dry years.

The Edge-tier result merits a specific interpretive note. The absence of a clean post-felling step at the Edge wells admits two readings: that the canopy-removal signal is genuinely confined to the felled compartment and does not extend to the surrounding edge, or that the Edge wells are undergoing a continuous decline — most plausibly the same site-wide β_1 decline described above, expressed at those wells — against which a step-change BACI model has little discriminating power. A step-change design tests for a discrete offset at a fixed date; it is structurally ill-suited to a trajectory that declines gradually from before the intervention onwards, as the Edge displacement series does under the record-length-balanced pre-felling window. The present record cannot separate these two readings, and the Edge result is therefore reported as a null with respect to a discrete clearfell step rather than as positive evidence of no effect.

The summer minima record makes this seasonal partition explicit. At WMC3 — the sole Impact well with sufficient pre-felling baseline — summer minimum depth was -1.573 m (mean, $n = 3$ years) before the April 2015 scraping, -1.620 m during the scraping era (2015–17, $n = 3$), and -1.600 m post-felling (2018–25, $n = 7$). The Forest Control tier was stable across the same period, with a tier-mean summer minimum of -1.822 m pre-scraping, -1.753 m during the scraping era and -1.766 m post-felling (Figure 33). Both Impact and Edge tiers track the Forest Control trajectory in summer extremes, with the felled wells' summer minima moving in parallel with unfelled forest while their mean monthly levels sit

above it. The climate-corrected spatial step-change mapping (Section 4.6.5; Figure 34) shows that the felled area resisted the site-wide drying trend by approximately 28 mm relative to the open dune.

The ANCOVA additionally includes an easting $\times$ time interaction term, which absorbs the progressive west-to-east deepening signal attributable to coastal erosion at the western boundary (Section 4.6.5). The felled compartment sits in the western half of the network, so without this correction the coastal erosion signal is partially attributed to the felled wells and offsets the clearfell recovery. With the easting $\times$ time term included, the ANCOVA recovers a clearfell-specific step of +120 mm at the Impact well — substantially larger than the uncorrected spatial map's +28 mm. The two estimates are not in disagreement: the spatial map and the ANCOVA differ in what they correct for, and the ANCOVA's additional correction is the one that isolates the canopy-removal signal from the coastal erosion signal.

The physical basis for this correction is set out in Section 4.9.4: the western-margin signal partitions spatially into an erosion-dominated band of net deepening along the dune-toe margin — coincident with the western C5 and C3 wells, in the western half of the BACI network — flanked by a sea-level-rise-dominated rise at the shore (Figure 54). The west-to-east form of the easting $\times$ time covariate is the net expression of this partition projected onto the network's east–west axis. Pye and Blott (2024) provide independent empirical corroboration of the underlying asymmetry: their dune-toe sequence records progradation in the north and chronic retreat in the south of the frontage — the same north-positive to south-negative gradient that the covariate absorbs. The linear easting term remains a first-order proxy, as Section 4.9.4 makes explicit: it captures the net west-to-east gradient but does not decompose it into the opposing erosion and sea-level-rise processes. Pye and Blott does not alter this; it grounds the proxy in a documented physical asymmetry rather than a purely statistical adjustment. The +28 mm figure should be read as a lower bound that retains the erosion confound; the ANCOVA figure is the clearfell-attributable component. The clearfell produced a significant relative recovery against the Forest control in the mean monthly signal; the response in summer extremes is set by the site-wide β_1 decline and the background climate trajectory, which operate symmetrically across felled and unfelled wells.

The 1990s-era attribution of water table depression to the forest (Cottingham, 1994; Bristow and Bailey, 2001; Betson et al., 2002) was conducted during the driest decade in the 95-year RAF Valley record (mean annual rainfall 793 mm against a long-run average of 853 mm; Section 4.1.1) and without the kind of climate-partitioned analysis applied in this study. The apparent depression of the 1989–1995 water table baseline is consistent with a compound signal of forest influence and exceptional climatic dryness that the data available at the time could not separate. The present three-counterfactual design separates these contributions and additionally partitions the canopy effect by season. The forest does suppress recharge — β_1 is lower under canopy, and the BACI record confirms a significant mean water table recovery following clearfell. The earlier studies would have recorded this winter-recharge

signal as a uniform water table depression; the present analysis shows that the canopy's effect on summer extremes is small relative to the site-wide β_1 decline operating across all tiers. The 1989–1995 baseline was therefore not solely a forest signal, nor solely a climate signal, but a compound of both — and a compound whose seasonal structure was not visible at the time.

The monitoring evidence therefore characterizes canopy removal as a winter-recharge intervention. Its effect on summer water table depth — the metric that determines dune slack viability — is set by the same site-wide β_1 decline that affects unfelled forest and open dune wells alike.

1.5.2 Forest Management Scenario Responses

The scenario framework (Section 4.10.2) quantifies the seasonal trade-off at the cluster level under the BACI-corrected β_2 multiplier ($\times 1.032$; Section 3.8). At C4, the clearfell scenario gives a net annual gain of +8.1 mm w.e./month, with the winter recharge gain exceeding the summer β_2 cost on the thin, high- β_2 substrate above the ridge. At C5, where the thicker substrate and lower β_2 attenuate the summer cost more strongly, the net annual gain is larger at +10.4 mm w.e./month. Fifty percent thinning produces a similar pattern at reduced magnitude (C4: +4.0, C5: +5.2 mm w.e./month). Both forest clusters respond in the same direction under clearfell and thinning, with C5 capturing roughly a third more recovery per unit area than C4. The cluster-level β_2 contrast (Section 5.3.1) — substrate as a major contributor, with depth-coupling and other factors also acting — translates here into per-area scenario response: the same canopy treatment produces different hydrological responses across the forest zone, and management prescriptions cannot assume uniform per-area benefit across the forest even where the underlying canopy is the same.

Broadleaf conversion produces a different seasonal balance. When the summer transpiration increase under deciduous canopy in full leaf is accounted for (β_2 summer multiplier $\times 1.075$, reflecting comparable growing-season transpiration rates between deciduous and Corsican pine canopies; Komatsu et al., 2011), the interception reduction from 24% to 15% during the recharge season is offset by the summer evapotranspiration penalty. The net annual volumetric effect is -0.7 mm w.e./month at C4 and $+1.6$ mm at C5 — near-neutral at C5 and slightly negative at C4 where the thin substrate amplifies the β_2 cost. The phenological advantage of winter leaf-off (zero interception during the recharge season) is offset by the growing-season transpiration demand, and the net annual effect is substantially smaller than climate-driven changes over the same horizon.

The three scenarios therefore differ in seasonal partition rather than in any one being categorically more or less effective than the others. Clearfell and thinning produce small positive net responses concentrated in the winter recharge season; broadleaf conversion produces a near-neutral annual balance that reflects the seasonal trade between winter recharge gain and summer transpiration demand. All three magnitudes are small relative to

the climate-driven Δh range projected over the same horizon (Section 5.7.1), and the cluster-level partition between C4 and C5 — substrate-related but not substrate-only — means that the same canopy treatment delivers a smaller per-area response at C4 than at C5.

1.6 Hydrological Effects of the Plantation

The Corsican pine plantation intercepts approximately 24% of incident rainfall before it reaches the aquifer (Freeman, 2008). This interception is a partition of the available PET energy budget, not an additional loss on top of PET (Section 3.7.2): intercepted water evaporates from wet canopy surfaces without stomatal control, consuming a fraction of the atmospheric energy that would otherwise drive transpiration and direct soil evaporation. The β_2 coefficient for the forested clusters therefore reflects the combined effect of this unmediated interception evaporation plus the residual stomatal transpiration and direct evaporation beneath the canopy. The interception fraction is well constrained by Freeman's (2008) site-specific measurement, and the β_2 contrast between the open-dune cluster (C3) and the coastal-forest cluster (C5) is consistent with a canopy interception loss of the order Freeman reports. That measurement is a single plot over a single season (August 2007 to January 2008), sited deep within the stand to minimise edge effects and assessed by Freeman as representative of the majority of the plantation in canopy cover, tree density and age. It is the only site-specific interception measurement available, and the quantities that depend on it — the interception-corrected WTF specific yields (Section 4.2.4) and the volumetric water balance (Table 4b) — inherit that limitation. The broader conifer plantation literature reports interception losses of 25–45% (Rutter et al., 1971; Gash, 1979; Gash et al., 1980), with the higher end deriving from dense Sitka spruce stands in UK upland catchments (Calder and Newson, 1979; Robinson and Dupeyrat, 2005) — a different canopy structure from the partially thinned Corsican pine at Newborough. Numerical modelling of comparable coastal pine settings in northern Poland produced interception losses of approximately 27% (Gumuła-Kawęcka et al., 2021, 2024), consistent with Freeman's figure.

The canopy has further hydrological effects beyond recharge suppression. It provides microclimatic shielding — reducing direct solar radiation, insulating the sand surface through the organic forest floor, and reducing wind-driven advective evaporation — and the forest floor and root-zone macroporosity produce a smoothed vertical recharge signal relative to the rapid, pulsed infiltration of the open dune (Calder, 2007). The mechanism by which that suppression operates has been measured at this site. Freeman (2008) recorded mean volumetric soil moisture beneath the Newborough plantation at roughly half that of the open Warren (approximately 6.5% against 17.7%; one-way ANOVA $p < 0.001$, all plot pairs separated at $p < 0.01$), with the depth structure inverted: the open-dune profile is wettest at 40 cm while the forest profile is wettest in the top 5 cm and dries downward. She attributes the contrast to capillary contact between the upper soil layers and the water table, maintained

in the open dune and lost beneath the canopy because the forest water table sits approximately 1.07 m deeper (ANOVA $p < 0.001$). The canopy therefore suppresses recharge twice over: once at the crown, where a fraction of rainfall is intercepted and re-evaporated, and again in the profile beneath it, where a deeper and drier unsaturated zone carries a larger storage deficit that each pulse must satisfy before the water table responds. Freeman's felled clearing plot, which carried a water table statistically indistinguishable from the open Warren, indicates that the second effect is a consequence of the canopy rather than an independent property of the forest zone. The lag diagnostic (Script 03) provides independent support for this smoothing effect. At lag-0, C5 Coastal Forest carries the lowest R^2 in the network (0.685, against 0.731 to 0.813 across the three open-dune clusters), indicating that the contemporaneous rainfall-to-water-table transfer is weakest beneath the coastal canopy; C4 Main Forest sits mid-range at 0.725, and the case for canopy smoothing there rests on the lag structure rather than on the lag-0 fit. At lag-1, the pattern inverts: C4's lag-1 R^2 (0.384) is the highest in the network, with a significant and physically coherent β_1 of 1.622 ($p < 10^{-12}$). No other cluster retains this level of explanatory power at one-month lag — C1 Lake Edge drops to $R^2 = 0.037$ (β_1 non-significant). The canopy delays and smooths the recharge pulse so that a substantial fraction of each month's rainfall reaches the forest water table during the following month rather than contemporaneously. The lag-0 formulation is retained for all clusters because it maximizes overall network R^2 and preserves the operational simplicity of a single-lag framework. The strong lag-1 signal at C4 indicates that the forest's lower apparent β_1 (2.487 vs 3.573–4.576 at the open dune clusters) partly reflects temporal redistribution of recharge rather than a reduction in total recharge reaching the water table. Freeman's dipwell record provides a direct observational analogue at the plot scale. Over her 2007–08 monitoring period the forest water table responded far less to rainfall than the clearing or open Warren, while the upper 40 cm of the forest soil profile responded more rapidly than either — the signature of a pulse arriving quickly at the surface and being absorbed within a thick unsaturated zone before it reaches the water table.

This temporal-redistribution reading carries a consequence for the management interpretation developed in Section 5.8.1. If the forest's lower apparent β_1 reflects recharge that is delayed rather than lost, then canopy removal recovers less water table gain than the open-dune-to-forest β_1 contrast would suggest — the intercepted-and-redistributed fraction was reaching the water table under canopy in any case, only later. This reinforces the P_flood finding (Section 5.8.1) that the forest clusters remain effectively unreachable regardless of canopy management: the winter recharge deficit that canopy removal recovers is smaller than the raw β_1 difference implies, and is in any event insufficient to lift summer minima sitting ~1.8 m below ground into the reachable band.

These effects, and their interactions with the aquifer substrate, are examined in the subsections below.

1.6.1 Atmospheric Draw, Specific Yield and the β_2 Coefficient

The compound nature of β_2 and the dominance of specific yield in the C4/C5 contrast were established in Section 5.3.1. The per-well analysis within the forest zone confirms and extends this finding.

C4 Main Forest has the highest β_2 in the network. This does not mean the forest water table experiences the greatest atmospheric demand — it means that a given evaporative flux produces the largest head change at C4, because C4 has the lowest effective specific yield. C4's low effective specific yield reduces the drainable volume per unit head change. The WTF-derived Sy estimates (Table 4c) give the measured contrast: C5's Sy is approximately a third higher than C4's. C4 overlies the bedrock ridge, which outcrops within the forest, so the nature of the underlying rock is established directly; what is not constrained without coring is the extent and depth of the ridge beneath the cluster, and hence the precise geometry of any buried ridges and the degree to which they impede lateral drainage. The contrast with C5 is consistent with C5 occupying deeper, more uniform coastal sand — essentially C3-type substrate beneath forest canopy, against C4's more C2-like setting — though the C5 sand depth is itself inferred from topography and coefficients rather than cored. The drainage decay half-life map (Section 4.9.3, Figure 49) is consistent with this: C4 wells show the longest $t_{1/2}$ values in the network (mean 40 months, range 16-88 months), indicating sluggish drainage that persists across years rather than free flow through a uniform sand column. The storage-drainage index $\tau = Sy/\beta_3$ provides a related perspective: where $t_{1/2}$ captures the pure drainage timescale from β_3 alone, τ additionally weights by specific yield, expressing the ratio of stored water volume to drainage flux at each well. Across the Newborough network, Sy and $1/\beta_3$ are spatially uncorrelated ($r = 0.13$), so the two indices carry independent information about aquifer architecture: $t_{1/2}$ reflects how fast the aquifer drains; τ reflects how much water it holds relative to that drainage rate. The near-identical spatial patterns of the two fields ($r = 0.98$ across the reference network) indicate that the drainage rate β_3 is the dominant control on both - storage capacity modulates the index but does not reorganize it spatially at the scale of this network.

C5 Coastal Forest has the second-lowest β_2 in the network despite carrying the same Corsican pine canopy as C4. This reflects C5's higher Sy — the thicker coastal sand provides more drainable storage per unit depth, so the same atmospheric demand produces a smaller head change. The per-well β_2 range within C4 alone spans nearly the full site-wide range (Section 4.9.5; Table 18), confirming that within-cluster substrate variation is at least as important as between-cluster canopy differences. Cluster-level centroid values summarize a spatially variable field; they should not be taken as implying uniform conditions across the forest.

Across the forest wells, elevation alone explains the bulk of the variance in β_2 (Section 4.9.5), with higher-elevation wells on the ridge showing systematically higher atmospheric draw sensitivity. The C4/C5 boundary corresponds to a clean elevation break between the

elevated ridge and plateau terrain and the low-lying coastal fringe, with no overlap. Both zones carry the same Corsican pine canopy, so the difference tracks elevation and water-table depth rather than canopy: C4 sits on the bedrock ridge (which outcrops within the forest) and C5 on the lower coastal ground, though the depth and extent of rock beneath the canopy are not constrained without coring. The implication for management interpretation is that β_2 values at any given forest well are dominated by position, not canopy: changes in β_2 following management must therefore be measured against a position-matched control if they are to be canopy-attributable. This is precisely the role of the Forest control tier in the clearfell ANCOVA (Section 4.6.2).

Although position — and the substrate inferred to underlie it — dominates the spatial pattern of β_2 , the post-clearfell coefficient analysis (Section 4.6.6; Table 11) shows that canopy shielding exerts a measurable effect on top of this baseline. β_2 increased at the intact-pine Forest Control wells following felling, consistent with the maturing canopy moderating the atmospheric demand signal reaching the water table. The response at the Edge tier was mixed, with no consistent direction across its four wells, and β_2 did not increase at the Impact well or at the Coastal Control. The shielding mechanism operates at the surface: intercepted water evaporates readily from wet canopy surfaces without stomatal control, consuming PET energy before it can act on the water table. Removing the canopy eliminates this interception pathway and exposes the water table directly to atmospheric forcing — a greater proportion of PET demand is then transmitted to the shallow aquifer through direct soil evaporation and grassland transpiration on the grazed clearfell surface. This shielding effect is superimposed on a position-driven β_2 baseline — reflecting S_y and the inferred substrate — that varies by a factor of two across the forest, and the cross-cluster β_2 pattern reflects S_y differences rather than canopy differences.

The depth-dependent PET diagnostic (Section 4.4; Section 3.7.1) is consistent with this interpretation. C1 Lake Edge shows the strongest depth coupling, consistent with shallow water-table evaporation near the lake being the dominant ET pathway, while C4 Main Forest shows the weakest depth coupling despite having the highest β_2 . If C4's high β_2 were driven by deep-root transpiration overriding capillary disconnection, C4 would show the strongest depth coupling, not the weakest. Together with the sunshine-hours residual diagnostic (Supplementary Note S6; Section 5.2), which returned a null result, this confirms that the standard SSM's fixed β_2 has already absorbed the bulk of the ET signal at the cluster level.

The practical consequence connects the plantation directly to the ecological outcome of most concern. Summer minimum water table depth is the dominant predictor of winter flooding potential in all western and forest blocks (Section 5.2; Table 12). Under a warming climate with increasing summer PET, any intervention that raises β_2 — including canopy removal — would amplify the head response to that PET, deepening summer minima per unit of evaporative demand and raising the rainfall threshold required for winter flooding. The post-clearfell β_2 increase at the intact-pine Forest Control wells is consistent with this mechanism.

The seasonal trade-off at the forest clusters — winter recharge gain from interception recovery against summer β_2 increase from loss of shielding — is quantified by the BACI-corrected scenario projections (Section 4.10.2; Section 5.5.2): net annual effects are small and positive at both C4 and C5, and substantially smaller than the climate-driven Δh range over the projection horizon. The BACI climate-sensitivity analysis (Figure 28; Section 4.6.2) provides a complementary line of evidence: the significant $CWB^2 \times \text{clearfell}$ interaction indicates that under the canopy the felling-attributable water table rise was concentrated in dry conditions, consistent with the canopy having partially buffered the water table against the worst dry-period deficits — a finding that is preliminary but operates on the relative scale that the amplification field (Section 5.3.2) cannot provide.

1.6.2 Summer Minimum Depths at the Forest Margin

The relatively shallow summer minima at NW2 and CEH2 are not explained by the water balance residual field. Neither well carries a positive residual (NW2 -0.0029 m/month, CEH2 -0.0048 m/month); both sit close to the network median, and no well in the network exceeds $+0.02$ m/month (Section 4.9.7, Figure 58). No local departure from closure, and so no unmodelled local input, need be invoked at these wells. Clearance of the plantation in this zone would eliminate the canopy contribution to the local water balance — the coefficient decomposition (Section 4.6.6) shows that β_2 increased at the intact-pine Forest Control wells following felling (mean $\Delta\beta_2 +0.145$, excluding the broadleaf well NW10), indicating that the canopy was moderating the atmospheric demand signal reaching the water table. Removing it at NW2/CEH2 would expose the water table directly to atmospheric forcing at a higher β_2 . The BACI analysis (Section 5.5) found that canopy removal in the adjacent clearfell zone raised the mean monthly water table but did not improve summer minima — the ecologically decisive metric — at the felled wells. The same seasonal partition would be expected at NW2/CEH2: clearance could lift the mean level while leaving the summer minimum, set by the site-wide β_1 decline, substantially unchanged.

The summer minimum depth map identifies where water table levels are currently most actively sustained against the background climate drying trend. The favourable position at the forest-adjacent margin reflects canopy condition together with local factors the monitoring record does not resolve; the available evidence does not support clearance as a route to preserving or improving it.

1.6.3 Broadleaf Conversion: The NW10 Natural Experiment

The monitoring network contains a direct observational test of broadleaf conversion effects. A block of mixed broadleaf woodland was established following clearfell in 1993 and restocked around 1996 on the north-western margin of the plantation, with dipwell NW10 situated within the block and monitored continuously from 2007. CEH2 and CEH34, both classified as C4 Forest under intact Corsican pine canopy, serve as the principal

comparators; NW11 and NW13 provide a paired pine-warren boundary transect approximately 90 m apart at virtually identical elevation.

The per-well SSM coefficients at NW10 are informative. Its recharge sensitivity $\beta_1 = 3.494$ substantially exceeds the C4 per-well median of 2.339 and the adjacent Corsican pine well CEH2 ($\beta_1 = 2.759$), consistent with the broadleaf canopy intercepting less rainfall than dense pine and allowing more recharge to reach the water table per millimetre of rainfall. The seasonal interception pattern expected from a deciduous canopy supports this reading: during the October–March recharge season the leafless broadleaf stand intercepts less rainfall than the year-round Corsican pine canopy. The meta-analytical review of Komatsu et al. (2011) reports that broadleaf deciduous forests in high-winter-precipitation climates consistently transmit more rainfall to the sub-canopy than coniferous forests at the same sites, with the effect most pronounced during the dormant season.

The atmospheric draw coefficient $\beta_2 = 2.628$ at NW10 is below CEH2 ($\beta_2 = 2.891$) and close to the C4 per-well median (2.634). The broadleaf stand does not show an increase in atmospheric draw above the C4 range. This is consistent with the seasonal offset of deciduous leaf-off during October–March partially compensating for higher growing-season ET demand, such that the year-round integrated atmospheric draw at the water table is comparable to that under adjacent pine. The comparison is complicated by positional differences: NW10 carries a higher β_3 (0.043) than the C4 median (0.019), so drains faster than most C4 wells, and its ground elevation and distance from the ridge differ from CEH2 and CEH34. These positional differences mean that the β_2 comparison isolates species effects imperfectly.

NW10's behaviour relative to its pine comparators differs depending on which comparators are used and whether the comparison is climate-normalized. When normalized to the shared climate signal, NW10's summer minimum sits consistently shallower than the pine interior composite (CEH2, CEH32, CEH33, CEH34), with a mean anomaly of +247 mm across the full record and +300 mm during the post-felling period (2018–2025). The anomaly is diminishing at –46 mm/yr (Section 5.6.3), consistent with the broadleaf stand progressively deepening summer minima as canopy closure increases interception and transpiration demand — the same direction of trajectory observed for the standing pine canopy as it matures. The direct comparison against CEH2 (NW10's nearest pine neighbour, 100 m downslope) shows NW10 deeper than CEH2 across the monitoring record; however, CEH2 itself shows a progressive deepening trend relative to the network mean that is not present in other pine interior wells (Section 5.6.3), which complicates the local-pair comparison. The reconciliation of these two pictures is the principal limitation of the NW10 experiment, and is addressed below.

The scenario framework (Section 4.10.2; Section 5.5.2) provides an independent quantitative reference. When the summer transpiration increase under mature deciduous canopy is accounted for (β_2 summer multiplier $\times 1.075$), broadleaf conversion produces a small net

annual effect — marginally negative at C4 (−0.7 mm w.e./month) and slightly positive at C5 (+1.6 mm). The convergence of two pieces of evidence — the diminishing climate-normalized anomaly at NW10 and the near-neutral scenario prediction at the forest clusters — points in the same direction, towards a broadleaf canopy whose annual hydrological effect at maturity is similar in net direction to the pine it replaced, even though the seasonal partition differs. The phenological winter recharge advantage at NW10 is real, but does not translate into a sustained shallower summer minimum on the available evidence.

NW10 is a single dipwell, and the positional confounders are of comparable magnitude to the observed contrasts. The spread between CEH2 (mean summer minimum −1.711 m) and CEH34 (−1.312 m), both Corsican pine wells within the same cluster, is 399 mm, exceeding the NW10 climate-normalized anomaly of +247 mm. No reliable attribution to canopy type is possible from a single dipwell, and a minimum of three to four instrumented wells distributed across the broadleaf compartment would be required before firm conclusions could be drawn. The 6–7 year extrapolation from the current −46 mm/yr trend should be treated similarly as an indicative trajectory rather than a quantitative projection, since the trend itself rests on a single well's post-felling record.

A related observation suggests the broadleaf block's hydrological footprint may extend beyond its physical boundary. CEH2, situated approximately 100 m downslope within the adjacent Corsican pine plantation, shows the progressive deepening trend relative to the network mean noted above (Section 5.6.3). This is consistent with the maturing broadleaf canopy progressively increasing winter recharge within the block, steepening the local hydraulic gradient and drawing down the water table in the adjacent downslope zone — the same mechanism that operates at the scraped slacks (Section 5.4). The observation is based on a single well comparison and cannot be attributed to canopy type with confidence, but it warrants targeted instrumentation downslope of the broadleaf block.

1.6.4 Stand Density, Thinning and the Dothistroma Natural Experiment

The preceding sections characterize the seasonal partitions produced by complete canopy removal and broadleaf conversion: clearfell produces a small positive net annual effect concentrated in the mean monthly water level, with the winter recharge gain offset by the summer atmospheric draw increase; broadleaf conversion produces a near-neutral annual balance where the phenological winter advantage is offset by growing-season transpiration. A third option — partial canopy retention through stand density management — has not been tested in the monitoring record but is mechanistically distinct from both alternatives.

No thinning experiment has been conducted within the study period and no thinned-stand wells exist in the network. The theoretical argument that an intermediate canopy density could retain a measure of microclimatic shielding whilst partially recovering the recharge penalty is mechanistically plausible — the β coefficient framework predicts that partial canopy retention would reduce β_2 less than complete removal while recovering a

proportional fraction of the interception loss from β_1 — but this remains a prediction rather than an observed outcome. The scenario framework (Section 4.10.2; Section 5.5.2) estimates that 50% thinning would produce approximately half the annual net response of full clearfell at both forest clusters, with a similar seasonal structure. The thinning and clearfell seasonal profiles are compared in Figure 71 (Section 5.7).

On prograding Atlantic shorelines, stabilized back dunes naturally succeed to deciduous woodland — the recognized climax community of the UK psammosere (Ranwell, 1972). The geomorphological position of the Newborough plantation is therefore not unusual for back-dune terrain; what differs from natural succession at this site is the species choice and the stand density, which reflect commercial forestry management. Freeman's (2008) site-specific measurement of 24% interception sits at the low end of the 25–45% range typically reported for UK conifer plantations, reflecting the exposed coastal setting and the partial thinning that has already occurred. The plantation's effect on the water balance reflects stand density as much as species choice.

Progressive *Dothistroma* needle blight infection is already driving natural canopy thinning across parts of the Newborough plantation. Monitoring the β_2 response of wells adjacent to blight-affected stands of varying severity would provide a direct field test of the canopy shielding mechanism quantified in this study, and would constitute a natural experiment in progress requiring no additional silvicultural intervention. Whether the trajectory leads toward a hydrologically distinguishable equilibrium cannot be asserted from the present data, but it is a well-defined monitoring question for the 2026–2030 period.

1.6.5 Canopy Salt Interception and Post-Felling Geochemistry

A further geochemical consequence of canopy removal appears to have received limited attention in the dune restoration literature but follows from the LAI-deposition relationship documented in coastal forest hydrology. Pine plantation canopies intercept sea salt aerosols with greater efficiency than open dune vegetation: throughfall studies in coastal *Pinus radiata* plantations have reported chloride deposition rates up to 89% higher beneath the canopy than in adjacent open fields, attributable to the high leaf area index of the needle canopy (Deng et al., 2013), and independent soil studies in the same species have shown systematically higher exchangeable sodium and magnesium pools under pine than under adjacent pasture (Giddens et al., 1997). The Newborough plantation is a different pine species (*Pinus nigra* var. *laricio*), so the specific deposition magnitudes do not transfer directly, but the qualitative mechanism — canopy interception of marine aerosols at LAI characteristic of mature coniferous stands — applies at any site exposed to onshore marine airflow. The Newborough Warren plantation has been exposed to frequent onshore Atlantic airflow throughout its 75-year history.

If this mechanism operates at Newborough at any meaningful magnitude, the forest soils will have accumulated an elevated Cl^- and Na^+ load relative to the open dune system over the

plantation's history. Forest removal simultaneously terminates this elevated deposition input and, by increasing aquifer responsiveness following canopy removal, accelerates the downward flushing of accumulated ions through the highly transmissive dune sand. The geochemical consequence is a progressive leaching of base cations from soils that previously received sustained buffering from canopy-concentrated deposition. Curreli et al. (2013) and Sival and Grootjans (1996) identify the maintenance of carbonate-rich groundwater buffering as critical for the persistence of basiphilous slack communities; this flushing dynamic represents a potential geochemical pathway by which large-scale clearance could alter the soil chemistry of base cation buffering, operating on a timescale longer than the water table response and potentially with a longer lag time before its effects become apparent in the vegetation.

A countervailing process should be acknowledged: pine needle litter produces organic acids that acidify the soil surface over time, and removal of the canopy would terminate this acidification pathway. The net geochemical effect of clearfell — salt-leaching versus relief from acidification — is likely to be site-specific and depth-dependent, and warrants targeted investigation through paired throughfall and soil porewater chemistry monitoring across the forest–dune boundary.

1.7 Spatial Groundwater Modelling: Findings and Limitations

A well-level scenario framework (Section 3.8; Section 4.9) was applied to translate the per-well SSM parameterization into site-wide summaries of equilibrium head change under prescribed climate and forest management perturbations. Climate scenario responses are spatially coherent and physically consistent with the per-cluster β coefficient structure: UKCP18 2050s forcing produces a site-wide mean annual Δh of -0.017 m/month, deepening to -0.030 m/month under UKCP18 2080s. The forested clusters carry larger annual declines than most open-dune wells; for C4 the mechanism is its low effective specific yield, which amplifies the head response to any change in atmospheric demand, while at C5 the larger annual signal reflects reduced net recharge under canopy interception combined with the cluster's already-steep observed trajectory (Section 4.8.1). These outputs are presented as indicative spatial summaries of the SSM equilibrium framework rather than calibrated predictions.

1.7.1 Seasonal Asymmetry under Climate Projections

The annual responses conceal a pronounced seasonal asymmetry that is ecologically critical (Section 4.10.1). Under both UKCP18 trajectories, winter Δh is positive at all clusters — increased winter rainfall under RCP8.5 Wales outpaces the concurrent winter PET increase — while summer Δh is strongly negative and carries the annual signal (site-wide means of -0.059 m/month at 2050s and -0.111 m/month at 2080s). This summer-dominated signature

reflects the same mechanism established in the observed record (Section 4.8.1): hydrological deterioration under climate change operates through intensified summer atmospheric demand rather than declining winter recharge. Annual-mean Δh therefore understates the ecological consequence of summer-dominated climate forcing; the seasonal partition should be reported alongside any annual summary.

1.7.2 Forest Management Scenarios and Structural Isolation

For forest management scenarios, the per-well framework carries an implicit structural feature that is itself informative: because β coefficient changes under canopy removal, thinning, or broadleaf conversion are applied only at C4 and C5 wells, the IDW-interpolated Δh field produces a response concentrated in the forest zone. The drawdown propagation analysis (Section 4.9.3, Figure 50) estimates a characteristic e-folding distance of 223 m, with the perturbation attenuating to negligible levels at all C1 and eastern C2 wells. Site-wide annual means under forest management are correspondingly small (Section 4.10.2; Section 5.5.2), because only 14 of 66 reference wells are in the two forest clusters and the propagation signal does not reach the eastern slacks at measurable magnitude.

C5 is the cluster where clearfell would yield the largest predicted summer minimum benefit (Section 5.5.2), driven by its higher effective specific yield and thicker substrate (Section 5.3.1). C5 is also the cluster with the steepest observed decline in the network — 3.9 times the summer rate of C1, with significant trends in both seasons and an ongoing collapse in seasonal amplitude (Section 4.8.1). Whether the predicted benefit could arrest the observed decline depends on the mechanism driving the decline: if the decline is primarily climatic, the scenario gain offsets several months of trajectory per year; if a structural process such as coastal boundary retreat contributes substantially, the gain offsets only the climatic component, with the structural component continuing. Two independent strands of the change analysis below indicate that the structural component is real and spatially concentrated at exactly these wells. The differential-movement map (Figure 63; Section 4.9.8), which removes the common climate signal and fits a per-well secular trend, identifies the south-western coastal and western margin — including the C5 coastal members — as the robust, window-independent sinkers on the site, consistent with the coastal-retreat gradient of Section 4.8.2 and the easting $\times$ time absorption of $-16.8 \pm 5.0 \text{ mm yr}^{-1}$ (Section 4.6.3). Independently, the two-window MSL5 change correlates with the 20-year summer-minimum slope at $r = 0.47$ ($p = 0.0009$, $n = 46$; Section 4.9.8), confirming that the coastal decline is a secular trend rather than a window artefact. The C5 decline therefore carries a structural coastal-retreat component that clearfell cannot offset, so the predicted scenario benefit should be read as addressing the climatic component only.

The spatial coefficient atlas (Section 4.9.2, Figure 48) provides independent characterization of the forest zone's hydrological position. The β_1 surface shows a sharp step at the forest boundary — recharge sensitivity drops from open-dune values to substantially lower values

beneath the canopy — and the β_3 surface shows an order-of-magnitude contrast between the fast-draining lake edge and the near-stagnant forest interior. These are not gradual transitions; they delineate a zone that runs its own local water balance with limited coupling to the open dune clusters. The bedrock ridge forms the northern boundary of the entire study area (Section 2), and all groundwater flows southward off the ridge toward the coast. The Darcy flow vectors (Section 4.9.6, Figure 57) show the forest zone discharging south-westward toward Caernarfon Bay, while the open dune discharges south-eastward toward the Menai Strait. The vectors are derived from the interpolated head surface and describe flow direction only; they are not evidence of a lateral input at the ridge margin, and the water-balance residual field does not support one — the forest clusters carry the most negative residuals in the network (C4 mean -0.0067 m/month), not a positive anomaly. A topographic high in the central dune field separates these two drainage paths. A canopy-management perturbation in the C4/C5 zone therefore propagates south-westward toward the coast, not south-eastward into the C1 and C2 clusters. Hydrological connectivity along the coastward gradient is empirically confirmed by the April 2015 scraping response at CEH36 (Section 5.4.2); this connectivity operates between the forest zone and the south-western coastal margin rather than between the forest zone and the eastern slacks.

1.7.3 Framework Limitations and Continuous-Flow Modelling

The conclusion that forest management perturbations do not reach the eastern slacks requires qualification. The forest drawdown propagation analysis (Section 4.9.3, Figure 50) estimates the steady-state head perturbation attributable to the existing Corsican pine canopy: the signal exceeds 50 mm only within ~ 100 m of the forest edge and is negligible (< 1 mm) at all C1 and eastern C2 wells. This quantifies the spatial reach of the canopy's current influence on the water table, and implies that removal of the canopy would produce a recovery of comparable spatial extent — confined to the forest zone and its immediate margins.

The dune-scrape and clearfell-baseline fields (Figures 55, 56) carry the same single-mechanism, steady-state caveats as the forest and coastal fields: the scrape edge magnitude is anchored to the one measured response, but the surrounding drawdown is inferred through the drain model rather than measured well-by-well, and the combined baseline field is a first-order superposition, not a closed budget. However, the Darcy flow vectors (Section 4.9.6, Figure 57) and the Pearson affinity analysis (Section 4.3) show that C4 and C5 are hydrologically connected along a continuous downslope drainage pathway. The clearfell's downslope effects on CEH36 (Section 5.4) demonstrate that perturbations in the C4 zone can propagate into C5. Whether they propagate further into C3 and C1/C2 at magnitudes below the propagation model's resolution cannot be resolved by the current framework, and would require a calibrated continuous-flow model.

The practical limits of the spatial framework lie in what it deliberately does not attempt. The per-well IDW framework is a geometric aggregator of SSM responses rather than a physical

model of groundwater flow, so it cannot represent cluster-to-cluster hydraulic coupling, lateral redistribution of localized recharge perturbations, or the diffusive response of the aquifer to boundary-flux changes. For the questions this study addresses — whether forest management in the C4/C5 zone affects summer minima in C1 and C2, and how climate perturbations partition across the cluster structure — the framework is adequate and the BACI record confirms its conclusions. Answering a different class of question, such as how a hypothetical intervention along the northern forest margin would propagate through the aquifer, would require a calibrated continuous-flow model, and for that the critical unknowns are hydraulic conductivity — currently constrained by a single tracer test ($K = 6.0$ m/day; Betson et al., 2002) — and aquifer thickness, inferred from four borehole logs across 700 ha (Bristow, 2003). Slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness are identified as the priority field measurements required to enable such a model. These data would also support a HYDRUS-1D vadose zone analysis (Šimůnek et al., 2008) at the cluster level, providing an independent test of the vegetation-driven recharge suppression signal identified in C4 and C5 (Gumula-Kawęcka et al., 2021).

1.7.4 Forest Scenario Predictions Against the Observed Record

The scenario framework's predicted seasonal responses can be tested against the observed post-felling record at WMC3 (Figure 71). Applying scenario-specific β coefficient adjustments to the observed C4 mean seasonal cycle under the BACI-corrected β_2 multiplier (Section 3.8) shows that the three management options diverge in summer response. Full clearfell and 50% thinning both produce shallower summer minima at C4, with clearfell recovering approximately 44 mm and thinning approximately 22 mm at the September trough relative to the pine baseline. These September-trough figures are seasonal-cycle head shifts and are not directly comparable with the equilibrium volumetric scenario bars of Section 4.5.6 (clearfell $\approx +13.5$ mm water-equivalent per month under summer forcing), which express a different quantity; the two are consistent in sign and in indicating a small forest-clearance gain. The clearfell and thinning gains arise because the interception recovery from canopy removal outweighs the β_2 increase at C4 in the summer equilibrium calculation, consistent with the small positive annual-mean effects reported in Section 5.5.2.

Broadleaf conversion produces the opposite seasonal response: the summer transpiration increase under mature deciduous canopy deepens the water table relative to the pine baseline through the summer window, with the largest departure — approximately 21 mm deeper — occurring in August. The reduced annual-mean interception under broadleaf delivers a visible winter recharge gain — the broadleaf line sits shallower than baseline from November through March — but this additional winter input drains through the highly transmissive sand before summer, and the growing-season transpiration penalty dominates the summer balance (Section 5.5.2). At WMC3, summer minimum depth shows no significant shift between pre-scraping and post-felling, consistent with the small predicted

clearfell effect at C4 and the BACI observed trajectory tracking close to the pine baseline through the summer window.

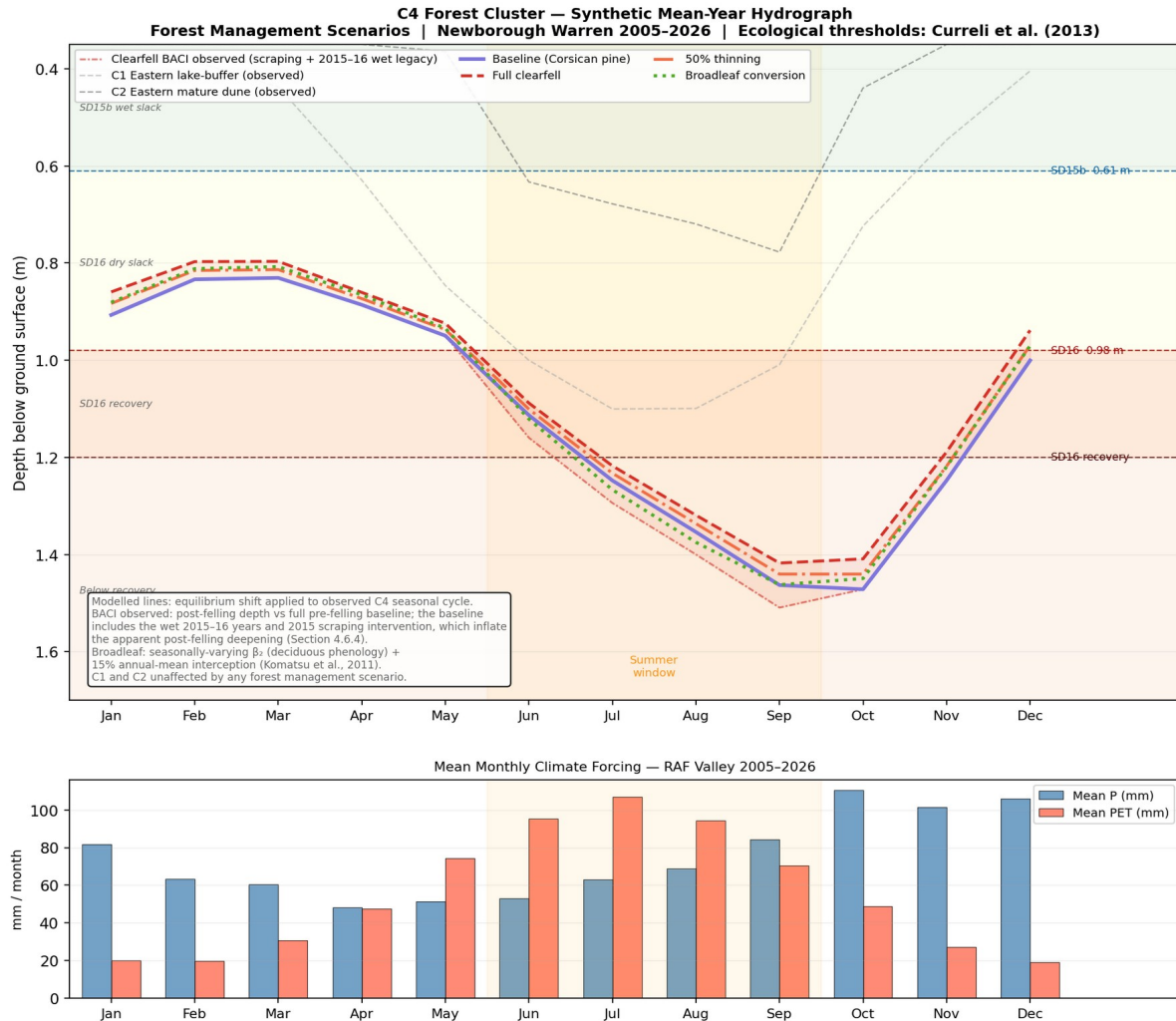


Figure 1.2: Synthetic mean-year hydrograph for the C4 Forest cluster under four management scenarios: baseline Corsican pine, full clearfell, 50% thinning, and broadleaf conversion. Lines show equilibrium head shifts applied to the observed 2005–2026 C4 seasonal cycle, derived from scenario-specific β_1 (effective rainfall, via canopy interception) and β_2 (ET draw, for clearfell and thinning) adjustments. The BACI-observed post-felling raw displacement at WMC3 (summer minimum -1.600 m post-felling vs -1.573 m pre-scraping baseline; Section 4.6.2) is shown for context — the gap between the modelled clearfell line and the observed record reflects the site-wide β_1 decline that suppresses summer minima across all tiers equally (Section 4.6.6), masking the significant mean monthly recovery detected by the ANCOVA ($+120$ mm Impact, $p < 0.001$; Section 4.6.2). C1 and C2 observed mean seasonal cycles are shown for ecological context; no management scenario produces any detectable effect on these clusters. Broadleaf conversion: seasonally-varying β_2 applied to reflect deciduous phenology (lower winter ET, higher summer ET, β_2 summer multiplier $\times 1.075$). Lower panel: mean monthly rainfall and PET from RAF Valley 2005–2026. Ecological zone colouring follows Curreli et al. (2013). Generated from 03_master_data.csv and 03_cluster_averages_maod.csv. (Source: 21_forestry_01_hydrograph.png).

1.7.5 Parametric Prediction and Observed Network Change

The parametric net state map (Figure 60, Section 4.9.8) and the observed MSL5 change map (Figure 62, Section 4.9.8) present the same aquifer through different lenses: one through the single-mechanism annual-mean driver fields of Sections 4.9.3–4.9.4, the other through the directly measured change in five-year mean spring water level between the pre-clearfell baseline (MSL5 window end 2017, springs 2013–2017, network mean –492 mm) and the current period (window end 2023, springs 2019–2023, network mean –589 mm). The comparison window is discussed below.

The standing canopy effect is invisible to a change map

The parametric map predicts a large net loss across the forest interior (–50 to over –150 mm) driven by the standing Corsican pine canopy drawdown. The observed MSL5 map shows a site-wide deepening with no positive zone in the forest interior. This is not a contradiction. The forest canopy drawdown represents a steady-state penalty relative to a hypothetical no-forest aquifer — a penalty equally present in both MSL5 windows. A before–after comparison cannot detect a driver whose effect is constant across both periods. The MSL5 comparison therefore cannot see the canopy drawdown at all; its magnitude is constrained by the parametric model (Section 4.9.3) and by the per-well β_2 coefficients of Section 4.9.2. The observed site-wide deepening reflects a change in the aquifer state between the two windows, not a response to the standing canopy.

What the MSL5 change map shows

The 2017–2023 MSL5 comparison (Figure 62; raw difference, $n = 59$ wells) shows the aquifer deeper across the site in the current window. The network mean MSL5 has deepened by 97 mm, from –492 mm (2017) to –589 mm (2023) below ground; 56 of the 59 wells exceed the ± 25 mm threshold and every one of them deepens — no well shallows. Because the figure differences two five-year windows, every time-invariant driver cancels — in particular the constant drawdown of the standing forest — so the surviving pattern reflects only drivers that progressed across the study period, and the deepening intensifies from the eastern dune interior toward the south-western coastal margin. The eastern C1/C2 dune wells deepen least (CEH25 –13, L7 –17, D7 –25, D8 –33 mm), several within the ± 25 mm near-neutral band. The clearfell Impact well WMC3 (–85 mm) and the scrape well CEH36 (–74 mm) deepen broadly in line with the –97 mm site-wide mean, showing no intervention-specific departure from the surrounding network. The most strongly negative zone is the south-western coastal margin: CEH22 (–229 mm), CEH21 (–194 mm), CEH18 (–158 mm), and the C5 coastal members CEH19, CEH17 and NW9 (–150 to –153 mm). The C4 Forest Control wells deepen by –130 mm (cluster mean), more than the site-wide average; as the standing-canopy drawdown cancels in a window-to-window difference and these are unfelled controls, this excess is neither a standing-canopy nor a clearfell effect but the

forested clusters' stronger differential response to the between-window climate anomaly, expressed in full because the map applies no climate correction (Section 4.9.8).

Window selection and the spring composition of the two windows

The choice of the 2023 current window requires justification. An alternative current window ending in 2025 (springs 2021–2025) was initially examined but not adopted: it contains the two wettest springs of the monitoring record, and the forested clusters respond more strongly to wet conditions than the open dune. In the 2025 window the Forest Control wells (CEH32, CEH34, CEH33, NW10, CEH2) show a large normalized gain of +107 mm, compared with +17 mm for the open-dune Climate Controls — an apparent Forest Control excess of +90 mm. Three checks indicate this excess arises from the spring composition of the two windows and from the limited resolution of a five-well group contrast, rather than from a real management or structural signal.

Check 1 — spring gap stability. The April–May gap between the Forest Control group mean and the Climate Control group mean was –520 mm pre-clearfell (2013–2017) and –534 mm post-clearfell (2018–2025, excluding 2024): a change of –13 mm, Mann–Whitney $p = 1.000$. The Forest Controls are structurally deeper than the Climate Controls in spring — reflecting the canopy drawdown — and this gap has not changed. In the 2025 MSL5 window the Forest Control excess over Climate Controls is +90 mm; in the 2023 window it collapses to –21 mm. The switch reflects which springs each window contains, not a change at the Forest Control wells. Measured against the 2013–2017 baseline gap, the five springs of the 2021–2025 window contribute +25, +42, –22, +28 and +16 mm respectively to the +90 mm excess. No single spring accounts for it: the largest contribution comes from 2022, and the anomalously wet 2024 spring contributes 28 mm, less than a third of the total. The 2019–2023 window contains 2019 and 2020, the two widest-gap springs of the record, which is why the same comparison there returns –21 mm.

The gap varies systematically with how wet the network is. Regressed on the site-mean spring level of the same year and of the preceding year, both terms are significant (contemporaneous +0.35 mm per mm, $p = 0.008$; one-year carry-over +0.24, $p = 0.022$; $R^2 = 0.57$, $n = 13$). This is consistent with the forested clusters' larger climate response (Section 4.9.8) acting together with their longer head-space recession time: $1/\beta_3$ is approximately 47 months at the Forest Controls against approximately 22 months at the Climate Controls (per-well group-mean β_3 ; the C4 and C3 cluster centroids give 49 and 17 months), so a spring anomaly retains about 78% of its amplitude after one year at the former against about 58% at the latter (Section 4.9.3).

This mechanism accounts for part of the excess but not all of it. Carrying the observed window wetness through the fitted relation predicts about +26 mm of the +90 mm. The remainder lies within the resolution of the contrast: the gap has an interannual standard deviation of 115 mm, so a five-year window mean carries a standard error of about 52 mm

and a difference between two such windows about 73 mm. On that footing the +90 mm excess is 1.2 standard errors and the -21 mm of the 2023 window is 0.3, neither separable from zero. This resolution limit is a property of a contrast between two five-well groups and does not bear on the network-mean deepening of Figure 62, which rests on 59 wells of which 56 exceed the threshold and none reverse. Both group figures are consistent with the unchanged pre- and post-clearfell gap established above, and neither supports a management or structural interpretation.

Check 2 — independent metric agreement. The MSL5 2017–2023 raw change correlates significantly with the long-run summer minimum slope from Script 25 (Pearson $r = 0.47$, $p = 0.0009$; Spearman $r = 0.43$, $p = 0.003$; $n = 46$ wells). These are entirely independent metrics: the summer minimum slope is a 20-year linear regression on annual July–September minima (Section 4.8.2, Script 25); the MSL5 comparison uses five-year April–May means from two specific windows. The convergence at $r \approx 0.47$ indicates that the 2017–2023 MSL5 map is showing the same spatial signal as the long-run secular trend established across the full monitoring record, not a window-specific artefact.

Check 3 — climate seasonality. Examination of monthly RAF Valley PET trends across the monitoring record reveals significant increases in May (slope $+0.066 \text{ mm month}^{-1} \text{ yr}^{-1}$, $p = 0.002$), June ($+0.047$, $p = 0.030$), and July ($+0.062$, $p = 0.006$). Total summer PET and summer rainfall show no significant trends ($p > 0.5$), but the intensification of early-summer atmospheric demand — occurring precisely when the water table is falling from its spring peak toward the summer minimum — provides an independent physical mechanism for the secular deepening visible in the MSL5 and summer minimum metrics. The rising early-summer PET draw is consistent with regional temperature increases during the growing season and applies uniformly across the site, producing the observed broad deepening with strongest expression at wells where the water table is most sensitive to atmospheric demand (C4, C5) and weakest at the lake-buffered eastern fringe (C1, eastern C2).

Together, these three checks indicate that the 2023 window is the defensible choice: its spring composition is not dominated by the wettest springs of the record, it matches the independent 20-year secular trend, and it is physically explained by the documented intensification of early-summer PET. The 2025 window was not selected because its dominant spatial feature, the Forest Control excess, is not separable from the spring composition of the window or from the resolution of a five-well group contrast.

Independence of spring readings within a window

The two-window comparison assumes the spring readings within each window are approximately independent, and the wide spread of drainage recession times across the network invites the expectation that this fails at the slow-draining wells. It does not: spring-level autocorrelation is approximately zero throughout (Section 4.9.8). Spring is the seasonal

maximum, reached after the winter recharge season, so each winter resets the water table and interannual recharge variability exceeds any residual carried forward; the monthly recession operates within the annual cycle rather than across it. The five-year average therefore behaves as intended, with an effective sample size of approximately five at every cluster. Window sensitivity is nonetheless real and graded across the network, but as a matter of amplitude rather than memory: C4 is the most window-sensitive cluster because its spring levels swing furthest, not because they are serially correlated, and what governs that amplitude is β_2 rather than β_3 (Section 4.9.8). Because the two parameters are correlated across wells, the recession time appears diagnostic when examined alone. This is the same higher β_2 that produces the forested clusters' stronger climate response, and it accounts for both the larger deepening at C4 and the greater sensitivity of any window comparison there.

Matched-window sensitivity and the secular trend

The three checks above establish that the 2017 → 2023 deepening is internally robust: it is not an artefact of window length or of gaps in the spring record, and it agrees with the independent 20-year summer-minimum trend. They do not, however, establish that the −97 mm headline is a measure of secular change. Because the amplitude of interannual variation differs so widely across the network, the magnitude of a two-window MSL5 difference — and at most wells its sign — depends on which pair of windows is differenced. Holding the well panel fixed across each comparison and admitting every well-sampled window pair (common panels of ≥ 40 wells, which excludes only the thin seven-well 2005–2009 baseline, so the spread reflects the window choice alone), the site-mean change spans −0.14 to +0.22 m (−136 to +221 mm) across the 66 admissible pairs and changes sign — 19 falling, 47 rising. The −97 mm 2017/2023 comparison is one interior cell; the most negative pairing is 2017 → 2020 (−136 mm) and the most positive 2015 → 2024 (+221 mm), the latter resting on the exceptionally wet 2024 spring, one of the two wettest of the record, whose effect on window composition the window-selection checks above already flag. That a single methodologically admissible change of window pair can move the site-mean estimate by more than a third of a metre, and reverse its sign, is why absolute site-wide change is not rested on the two-window comparison (Script 34; the window-sensitivity figure for this section). Consistent with this, a trend fitted to the site-mean spring level across the full record is -7.0 mm yr^{-1} (AR-corrected, $p = 0.52$; the figure's own-panel ordinary-least-squares variant is -8.0 mm yr^{-1} , $p = 0.42$), not significantly different from zero. The two-window MSL5 comparison is therefore best read as a snapshot of where the aquifer stood in one window relative to another, not as a secular rate; the window-independent secular signal is recovered instead by the differential-movement and envelope analyses that follow (Figures 62 and 64).

Absence of a detectable management signal in the MSL5 map

The MSL5 2017–2023 map shows no spatially distinct signal attributable to either the 2015 dune scraping or the 2017 clearfell. WMC3 (clearfell Impact) shows −85 mm and CEH36

(scrape well) shows -74 mm — both deepening broadly in line with the -97 mm site-wide mean and indistinguishable from the surrounding network. The Forest Controls show -130 mm — deeper than baseline, consistent with the site-wide secular trend and not exhibiting any departure from it that could be attributed to management. This is consistent with the BACI finding of Section 4.6.4 that the clearfell recovery is expressed in the monthly-mean water level but not in five-year spring means at the Impact or Forest Control wells. The scraping benefit at CEH36 ($+129$ mm, Section 4.5.2) is expressed in the monthly-mean era comparison but is largely absorbed by the construction of the five-year spring means. The 2013–2017 baseline window already contains two fully post-scraping springs (2016 and 2017), with the 2015 spring straddling the April 2015 works, so much of the benefit is present in the baseline as well as in the current window and does not survive the difference. CEH36's -74 mm against the -97 mm site-wide mean is consistent with a small residual of the benefit persisting. This is a window-overlap effect rather than a drainage one: the scrape permanently lowered the ground surface to a new equilibrium and is not a transient anomaly subject to head-space recession.

The coastal decline signal

The coastal margin decline in Figure 62 is the spatially distinctive element of the MSL5 map that is not explained by the site-wide secular deepening alone. The south-western coastal wells (CEH22 -229 mm, CEH21 -194 mm, CEH18 -158 mm; C5 coastal members CEH19, CEH17 and NW9 -150 to -153 mm) show substantially larger deepening than the network mean, consistent with the coastal-retreat gradient documented in Section 4.8.2 and with the BACI easting $\times$ time absorption of -16.8 ± 5.0 mm yr $^{-1}$ (Section 4.6.3). The parametric map places these wells in a modest net-gain zone (SLR marginally exceeding the Storm Brendan erosion drawdown), which does not match the observed decline. This discrepancy is consistent with the known limitations of the one-dimensional erfc SLR model at a finite bounded dune body and with the residual unexplained easting-correlated decline identified in Section 4.6.3 (Table 8).

Differential movement with the climate signal removed (Figure 63)

The differential-movement map (Figure 63, Section 4.9.8) recovers the secular component that a two-window comparison cannot isolate. By removing the common climate signal shared across the network and fitting a per-well trend in spring (MAM) level, it shows which parts of the aquifer are moving relative to the site as a whole, independently of how favourable any individual spring happened to be. Over 2011–2025, 8 of 74 mapped wells move significantly, and the significant movers are almost entirely sinkers on the south-western coastal and western margin — CEH22 (-26.5 mm yr $^{-1}$), CEH21 (-15.1 mm yr $^{-1}$) and CEH11 (-13.3 mm yr $^{-1}$) among them. Extending the window to 2005–2025 firms the pattern to 13 of 77 wells, with the coastal and western sinkers (CEH22, NW8, NW9, CEH17, CEH3) strengthening and remaining the dominant significant signal. The coastal-margin decline identified above is thus confirmed as the robust, window-independent secular

movement on the site, consistent with the coastal-retreat gradient of Section 4.8.2 and the easting $\times$ time absorption of Section 4.6.3.

Reconciling the parametric forest loss (Figure 60) with the observed forest holding (Figure 63 and Figure 64)

The forest interior behaves very differently in Figure 63 from the way the parametric net-state map (Figure 60) would lead one to expect. Figure 60 predicts a large net loss across the forest interior under the standing-canopy drawdown; in Figure 63 the forest, far from declining, holds its position relative to the rest of the Warren, and the forest control wells (CEH32 +20.5, CEH34 +17.2, CEH2 +14.9 mm yr⁻¹) are among the few that drift upward. This is not a contradiction between the two figures; they describe different quantities. Figure 60 is a steady-state penalty — the depression the standing canopy maintains relative to a hypothetical no-forest aquifer, a constant offset present throughout the record — whereas Figure 63 is a rate of change with the common climate signal removed, and a constant offset, however large, contributes nothing to a secular trend.

The absolute climate-removed map (Figure 64) confirms this reading from the other direction. Where Figure 63 shows the forest holding its position *relative to the network*, Figure 64 — which removes the climate signal without re-referencing to the site mean — shows the same wells close to zero in *absolute* terms: the main-forest cluster sits at C4 $\approx$ +2.7 mm yr⁻¹, alongside the lake-edge and dune clusters (C1 $\approx$ +2.1, C2 $\approx$ +4.6 mm yr⁻¹), none of them trending materially in either direction once the referencing is removed. The upward drift of the forest wells in Figure 63 is therefore a property of the reference frame — the forest declining more slowly than a wet-spring-lifted network mean — and not a secular gain. Read together, the relative and absolute maps agree: the forest interior is stable, neither recovering nor declining, while the genuine secular losses are confined to the western and coastal margins (C3 $\approx$ -3.4, C5 $\approx$ -11.6 mm yr⁻¹).

The forest's predicted sensitivity is nonetheless real; it is expressed in two ways that both leave its relative position intact rather than as secular decline. First, as climate volatility: the amplification field (Figure 65) shows the forest interior magnifying the dry-to-wet spring swing more strongly than any other zone — the C4 forest at $\approx 1.7\times$ the network-mean swing of 0.75 m, against $\approx 0.6\times$ at the lake edge and $\approx 1\times$ on the open dune — so the forest moves furthest within each climate cycle while not trending relative to the site across cycles. Second, through its position in the recession geometry: the forest occupies a high, slow-draining mound (C4 head-space recession time $1/\beta_3 \approx 49$ months against ≈ 15 months at open-dune C2), and a slow-draining high holds its relative elevation as the whole network rises and falls. The amplification field correlates with the per-well β_2 coefficient ($r \approx 0.8$), a marker of this deep, slow-recession store rather than a literal evaporation rate, since β_2 measures head response per unit atmospheric demand and is therefore raised where drainable storage is small (Section 4.9.2). The parametric forest loss and the observed forest holding are therefore two consistent readings of the same aquifer — annual-mean steady-state

drawdown versus climate-removed secular drift — not a discrepancy to be resolved in favour of one figure.

The dry-year spring envelope and slack viability (Figure 66)

The envelope analysis also bears directly on ecological condition. Figure 66 maps the spring water-table depth in the dry extreme years (2011, 2012, 2019) against the Curreli et al. (2013) wet- and dry-slack thresholds (SD15b 0.61 m; SD16 0.98 m). Across the forest interior and the western residual block the dry-year spring table already sits below the SD16 dry-slack threshold. Because spring is the seasonal high, a spring level already below a summer-minimum threshold guarantees that the summer minimum is deeper still: the contour is a strict, conservative lower bound on the area below dry-slack viability, and the true sub-threshold footprint in late summer of a dry year is larger. This is the dry-year complement to the secular picture — the coastal margin is where the table is declining fastest (Figures 62 and 64, the latter confirming the coastal decline in absolute terms), while the forest and western dune are where, in dry years, the table is already deepest relative to the ecological thresholds (Figure 66). The single raised inter-slack well CEH10 is excluded from the interpolated surface, as it measures slack-edge rather than slack-floor ground; it is retained as a distinct slack-edge marker.

Robustness of the BACI clearfell estimates

The finding that no management signal is detectable in the MSL5 comparison bears on the interpretation of the BACI estimates, and it is important to distinguish which estimates are robust and which require qualification.

The primary ANCOVA-BACI monthly-mean step change at WMC3 relative to the Forest Control counterfactual (+120 mm, $p < 0.001$; Section 4.6.2) is robust. It is derived from the continuous monthly record using a common-climate covariate structure, controls for the shared climate trajectory of Impact and Control tiers through the ANCOVA design, and its significance is large enough to survive substantial uncertainty in the counterfactual. The directional conclusion that the clearfell produced a real monthly-mean water-table recovery at WMC3 is well-supported and is not affected by the MSL5 window analysis.

The era-split β coefficient shifts ($\Delta\beta_2$, $\Delta\beta_3$) at the Forest Control tier (Section 4.6.6) are less robust. The network-wide α decline post-felling, visible even at the open-dune Climate Controls (NW7 -27 mm month $^{-1}$, NW5 -23 mm month $^{-1}$), indicates that the two eras have different climate-adjusted baselines driven by the changed antecedent storage state that the monthly SSM does not capture. The post-felling β_3 decline at the Forest Controls may therefore partly reflect this era-level climate confound rather than a real physical change in the aquifer, and should be treated as indicative pending corroboration from direct hydraulic measurements.

The summer minimum analysis (Section 4.6.4), the four-zone BACI corroboration (Section 4.6.7), and the BACI-corrected β_2 multiplier used in the scenario framework (Section 3.8.4)

are derived from the full monthly record with the pre-specified BACI design and are unaffected by the MSL5 window issue. The Edge-tier β_2 increase is consistent across the ANCOVA, mixed-model and SSM residual analyses and is considered robust.

Priority measurements to reduce uncertainty

Three field measurements would materially resolve the remaining uncertainties. First, continued monitoring through a dry spring sequence equivalent to 2013–2017 would test whether the MSL5 pattern under neutral climate conditions shows any management signal above the secular deepening trend. Second, slug tests at two or three representative C4 wells (CEH32, CEH34, CEH13) would provide hydraulic conductivity estimates independent of the SSM coefficient fits, allowing the post-felling β_3 decline to be evaluated against measured drainage properties. Third, a ground-penetrating radar survey of aquifer thickness, with coring, would replace the indirect β_1 -proxy constraints with direct measurements of thickness and specific yield, and would establish whether a ridge boundary flux exists at all rather than assuming one.

1.7.6 External Validation Against Dune-Slack Vegetation, and a Structural Wetness Index

MSL5 (Section 4.8.4) follows van Willegen et al. (2025), the strongest single hydrological correlate of dune-slack vegetation moisture at Newborough. As implemented here it reproduces their published spring levels to within ~11 mm and predicts their co-located Ellenberg-F between wells ($r = +0.83$, $n = 18$), confirming the metric carries the ecohydrological signal attributed to it (Section 4.8.5).

The state-space coefficients afford a structurally-derived estimate of the same quantity — an equilibrium water-table level obtained by inverting the head-dependent drainage term under long-term mean climate. Tested directly against Ellenberg-F it is statistically indistinguishable from MSL5 (RMSE 0.35 vs 0.34 Ellenberg-F units; overlapping confidence intervals; Williams $p = 0.81$), and the two distribute almost identically across per-well accuracy bands. That coefficients calibrated purely on dynamics predict vegetation moisture as well as the direct spring-level metric is the substantive result; on $n = 18$ it is equivalence within the data, not proven identity.

Calibrated onto the MSL5 scale, the index also reconstructs observed MSL5 across the open dune (clusters C1–C3) to 119 mm RMSE, generalizing to the wells van Willegen never studied (100 mm) — but it is scoped to the open dune, as MSL5 itself is (Section 4.8.4): in the forest the coefficients are least constrained (Section 4.9.2) and prediction degrades to ~227 mm, too coarse to classify a well across a Curreli threshold.

Its practical value should be stated precisely. The index is not simpler to construct than MSL5 — it additionally requires climate data and a fitted model, and it is fragile where the fit is weak. Its one genuine advantage is record length: an SSM fit needs of order thirty

months, whereas MSL5 needs five clean consecutive springs, so the index is available earlier in a monitoring programme. At Newborough, with a 21-year record, that advantage is immaterial; its utility is transferability to other dune systems or to early monitoring elsewhere, where MSL5 cannot yet be formed. A distinction matters here. The agreement with vegetation is independent of MSL5: the equilibrium index is built from the state-space coefficients and climate alone, so its prediction of Ellenberg-F is an external ecological validation of the model, not a restatement of the spring-level metric. The reconstruction of MSL5 from the index, by contrast, is calibrated on MSL5 and is a diagnostic of usability rather than an independent finding. The vegetation result rests on eighteen wells at a single site, so its statistical power is limited, but it stands on its own — coefficients fitted purely on month-to-month dynamics carry an ecological signal not borrowed from MSL5.

1.8 The Reach of Forest Management Interventions

The preceding sections have characterized the clearfell signal as concentrated in the mean monthly water level rather than in summer minima at the felled wells (Section 5.5), with the canopy acting both as a recharge-suppression mechanism through interception and as a microclimatic buffer on atmospheric demand at the water table (Section 5.6). A separate question is whether forest management in the C4/C5 zone could in principle improve water table conditions in the open dune clusters — C1 Lake Edge, C2 Dune and C3 Western Residual — where the ecological thresholds are most critical and the remaining wet-slack resource is concentrated.

The available evidence indicates that forest management in the C4/C5 zone does not measurably affect water table conditions in C1, C2 or C3. The spatial scenario framework (Section 4.10.2) produces no detectable Δh response at C1, C2 or C3 under any forest management scenario. The climate-corrected spatial step-change maps (Section 4.6.5; Figure 34) show that the felled area moved less than the site-wide drying trend — consistent with the significant ANCOVA result (Section 4.6.2) — but this effect was confined to the forest zone and did not propagate into the open dune clusters. Site-wide annual means under forest management are correspondingly small (Section 5.7.2), because only 14 of 66 reference wells are in the two forest clusters and the framework applies no spatial propagation from the forest zone into other clusters. Broadleaf conversion is near-neutral at the forest clusters themselves (Section 5.5.2) and produces no meaningful site-wide signal. The BACI monitoring record is consistent with this structural isolation: wells outside the forest zone showed no detectable felling-specific response once common-mode climate variability was accounted for (Section 4.6.7). The transect from uncleared plantation interior to clearfell core showed spatially uniform post-felling trajectories with no significant distance gradient (Section 4.6.5; Figure 36), indicating that the recharge recovery signal is real but local.

This structural isolation is not a framework artefact. The per-well analysis (Section 5.6.1) shows that the C4/C5 boundary corresponds to an elevation transition (and inferred substrate

change), and the drainage geography indicates that forest-zone perturbations propagate south-westward toward Caernarfon Bay rather than eastward into the C1 and C2 clusters (Section 5.7.2). The forest drawdown propagation analysis (Section 4.9.3, Figure 50) provides a quantitative upper bound: the steady-state head perturbation from the existing 75-year-old plantation is negligible at all C1 and eastern C2 wells, despite the plantation having had its full lifetime to reach equilibrium. Any recovery from clearfell would propagate outward on the same multi-year to multi-decadal timescale (Section 5.4.3), and the equilibrium recovery footprint would be spatially comparable. The P_{flood} equations (Table 15; Section 4.7.5) characterize the consequence at the forest clusters themselves: at Forest cluster summer minima of approximately 1.8 m below ground, the required winter rainfall to achieve flooding exceeds anything in the 95-year RAF Valley record regardless of canopy management.

The clusters where the ecological stakes are highest — C1 and C2, whose summer minima sit within 0.2–0.4 m of the SD15b and SD16 thresholds — show no detectable response to forest management in the C4/C5 zone, and their summer minimum trajectory is driven by the site-wide climate signal. Winter flooding in the Western Residual is controlled almost entirely by antecedent summer minimum depth (Table 12); winter rainfall amount is non-significant in C3, C4 and C5, and secondary even in C1 and C2 (Section 5.2.5). The primary restoration lever is therefore summer minimum depth, over which forest management in the C4/C5 zone has no detectable influence at C1 or C2.

1.8.1 P_{flood} Recoverability and the Operational Boundary

The P_{flood} threshold mapping (Section 4.7.4, Figures 37-39) characterizes the recoverability of each well in the network under realistic rainfall conditions. Of the 88 classified wells, 59 (67%) are reachable under mildly wet winters ($1.0 \leq \lambda < 1.5$) — these are concentrated in C1 and C2 and represent the zone where targeted scraping can shift a marginal well into reach of a mildly wet winter. A further 24 wells (27%) require wet winters of varying rarity ($1.5 \leq \lambda < 2.5$), and the remaining 4 wells (5%) are effectively unreachable under the present climate ($\lambda \geq 2.5$). The unreachable wells are concentrated in the forest zone — three in C5 Coastal Forest (CEH17, FE3 and CEH31) and one in C4 Main Forest (CEH33) — where summer minima below approximately 1.8 m below ground place the required winter rainfall beyond anything in the 95-year RAF Valley record (the wettest observed winter, 745 mm in 2001, delivered only $1.51\times$ the climatological mean and 60% of the $\lambda = 2.5$ demand). C3 Western Residual occupies the middle ground — most C3 wells sit within or just above the wet-winter-reachable band (median $\lambda = 1.27$) but the deeper-end wells approach $\lambda = 1.9$ (with the deepest at $\lambda = 1.97$), and C3's near-zero winter flooding record (Section 5.1) indicates that even nominally reachable thresholds are rarely achieved in practice. The recoverability map delineates a clear operational boundary: the target zone for direct intervention is C1/C2 where λ values cluster close to 1.0 and a single scraping event can bring a well within reach of a mildly wet winter.

The storage–drainage index $\tau = Sy/\beta_3$ adds a further dimension to the achievability picture that λ alone does not capture. Two wells with identical λ can differ substantially in their sensitivity to the distribution of rainfall within the winter. A well with a short recession time (fast-draining, typical of C1) has $t_R = 1/\beta_3 \approx 8\text{--}10$ months: significant drainage losses occur between monthly recharge pulses, so an early wet October contributes relatively little more to the February peak than a wet February. A well with a long recession time (sluggish, typical of C4) has t_R exceeding 50 months: drainage losses within a single winter are negligible, and the aquifer integrates rainfall contributions almost equally regardless of when in the winter they arrive. The operational implication is that at long-recession wells sitting near a λ threshold, the achievability classification is more stable — a borderline $\lambda = 1.4$ well in C4 will flood given any sufficiently wet winter, with the exact timing of rainfall within the season mattering little. At short-recession wells the same λ carries greater sensitivity to whether the wet months fall early or late in the recharge window. This distinction is invisible in the λ map but is encoded in the recession-time field (Section 4.9.3) and should inform operational interpretation of achievability classifications near the $\lambda = 1.0$ and $\lambda = 1.5$ boundaries.

1.8.2 Climate Trajectory as the Binding Constraint

The clearfell produced a significant mean water table recovery against the Forest control (Section 5.5), but the recovery is concentrated in the mean monthly water level rather than in summer extremes, and the site-wide β_1 decline driving summer minimum deterioration appears in both felled and unfelled tiers (Section 5.5.1). The seasonal pattern at Newborough places the operational weight of canopy management on the winter recharge season, while the ecologically critical constraint — summer minimum depth — is set primarily by the climate trajectory operating across all clusters regardless of canopy cover.

Whether this seasonal asymmetry generalises to other coastal dune sites depends on the local water balance. The seasonal prediction equations at Newborough show that summer minimum depth dominates winter flooding prediction in the western and forest blocks, while winter rainfall is the dominant predictor only at the eastern lake-buffered and shallow till clusters (Table 12). Where this pattern holds at other sites, canopy management would address a different season from the ecologically critical one. Where the local water balance is winter-rainfall-limited rather than summer-minimum-limited, the calculation would differ. The Newborough finding cannot be directly transferred without site-specific seasonal partitioning of the flooding-control mechanism.

At Newborough specifically, the progressive redistribution of rainfall toward wetter winters and drier summers under a changing climate is visible in the declining summer minimum trends (Section 4.8.1). The corrected scenario projections (Section 4.10.2; Section 5.5.2) predict small positive net responses at the forest clusters under clearfell and thinning, and near-neutral or slightly negative responses under broadleaf conversion. These magnitudes

are substantially smaller than the projected climate-driven changes over the same horizon (Section 5.7.1). Forest management perturbations do not propagate to the C1/C2 zone where the ecological need is greatest (Section 5.8). The operational domain for direct hydrological intervention at Newborough therefore lies in topographic scraping in the C1/C2 zone where the P_flood threshold remains achievable under realistic winter rainfall (Section 5.8.1).

These reaches can be consolidated on a common footing by expressing each intervention and driver as an equilibrium head change against distance from its source (Figure 72). The scrape dipole and the forest drawdown, already discussed, decay exponentially over $\lambda \approx 230$ m and are spatially confined; the figure adds the two comparisons the preceding analysis has not made — coastal retreat and climate. A single Storm-Brendan-class 6 m retreat imposes only a shallow edge drawdown (≈ -21 mm), but because the coastal signal decays linearly across the whole western dune field rather than exponentially, it reaches far further inland (to $L \approx 894$ m) than any management effect. Sustained over five years, the fitted coast-edge retreat trend accumulates to a drawdown comparable in magnitude to the standing forest yet spread across the entire western system. The climate term, by contrast, has no reach decay at all: it acts uniformly across the site, modest at any one point but everywhere, and on the shared five-year footing the accumulating coastal curve crosses it at roughly 700 m — coastal retreat is the larger driver along the western fringe, while the climate decline, small but ubiquitous, carries the larger site-wide mean. The single measured off-site scrape point (WMC3, ≈ -55 mm at 262 m, reproduced across the 2015 and 2023 scrapes) anchors the scrape dipole, whose wider cone remains a modelled projection. Read with the caveat that every curve is a single-mechanism steady-state construction anchored at one or two measured points, and that the coastal and climate curves are forward projections of a fitted trend rather than observations, the figure reinforces the section's central finding: the management interventions most often proposed are spatially localized, whereas the shoreline retreat and the climate decline are basin-wide — The mechanisms behind the four drivers are drawn schematically in Figure 73.

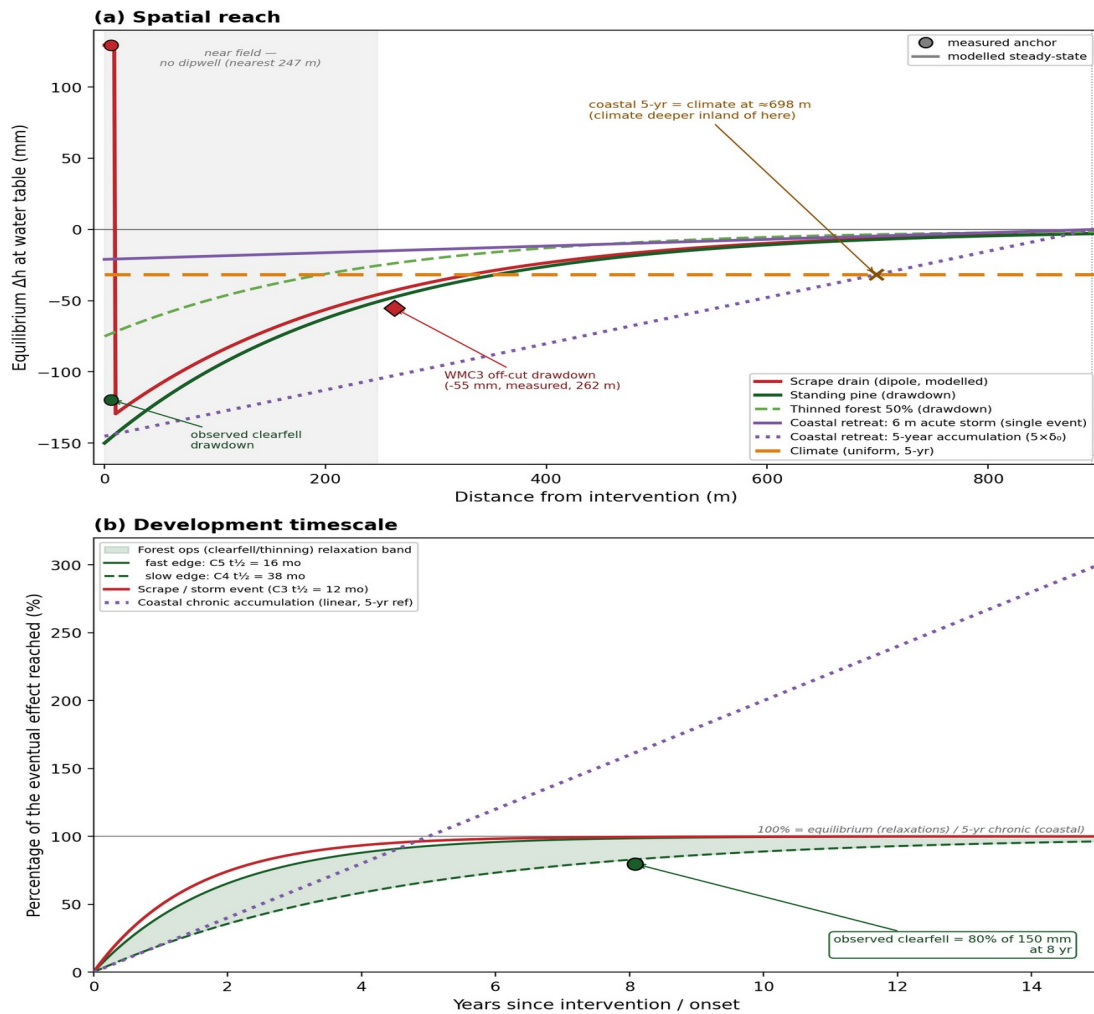


Figure 1.3: Spatial reach and development timescale of the management interventions, coastal retreat and climate. **(a) Modelled equilibrium water-table displacement change (Δh)** against distance from each source. The scrape dipole and the standing/50%-thinned forest drawdowns decay over the open-dune e-folding length ($\lambda \approx 230$ m); the two coastal curves show a single 6 m storm event and a five-year accumulation of the fitted coast-edge trend ($5 \times \delta_0 \approx 145$ mm at the toe, $\delta_0 = 29$ mm yr^{-1} , reach $L \approx 894$ m); the climate term, being spatially uniform, is drawn flat at its five-year amplitude (≈ 32 mm). The coastal and climate accumulations cross at ≈ 698 m (marked $\times$). Measured anchors are the CEH36 scrape and clearfell BACI responses at the source ($r \approx 0$) and the reproducible WMC3 off-cut drawdown (≈ -55 mm at 262 m), the one evidenced off-site point; the wider scrape cone, the coastal curves and the climate line are modelled. The shaded band marks the near field the network cannot resolve (nearest well 247 m). **(b) Time to develop**, as a percentage of the eventual effect; being dimensionless it composes with panel (a) — the head at a given time and distance is the panel-(b) fraction times the panel-(a) magnitude. Forest operations relax as a saturating exponential spanning the forest-cluster half-lives (C5 ≈ 15 , C4 ≈ 34 months); a scrape or single storm-retreat step relaxes on the faster Western Residual (C3) half-life of about 12 months, sharing one curve; the coastal decline instead accumulates roughly linearly toward no equilibrium, the saturating-versus-linear contrast distinguishing a management relaxation from ongoing accumulation. The observed clearfell response (≈ 120 mm, about 80% of the 150 mm modelled equilibrium, at roughly eight years) falls on the forest band, indicating the modelled magnitudes are consistent with the record once the timescale is applied. Drainage coefficients and λ are live pipeline values; the coastal, storm, thinning and climate curves are modelled scenarios flagged as such. (Source 09f_management_effects.png).

The four drivers act on the water table by distinct mechanisms, drawn schematically in Figure 73. A **dune scrape** cuts down to meet the water table, so the excavation becomes a pool at the unchanged local level; cutting headward into the landward edge of a slack extends the low seaward control inland and draws the adjacent slack below its floor. Both effects are measured: the cut wets by +129 mm (CEH36 against its control) — the pool sitting marginally higher because the headward cut reaches ground where the inland head is naturally higher, not because excavation lifted the table — while the near-field well WMC3 falls by 55 mm in 2015 and 54 mm in 2023, a reproducible drainage-capture drawdown across two independent scrapes. This is the one evidenced off-site point; there is no network-wide drawdown cone (Section 4.9.2). The scrape is locally wetting but redistributive, with a small net site-wide loss, and it relaxes to a new level over a few years.

A **clearfell** lifts the canopy's local suppression of the water table (about 150 mm under standing Corsican pine, whose canopy intercepts roughly a quarter of incident rainfall), and the felled slack recovers toward the no-tree table. The measured recovery is +0.12 m as an annual mean (significant), but in summer — the ecologically limiting season — the step is only about +0.05 m and is not statistically significant, so felling does not restore the summer minimum. The recovery settles over one and a half to three years.

Coastal retreat lowers the seaward boundary as the shoreline recedes (3–8 m per year), drawing the water table down over a coastal fringe. Unlike the other three, this drawdown is not directly measured: the retreat rate is known but its effect on the water table is modelled at about –29 mm per year at the shore ($\approx$ –580 mm over twenty years), tapering inland over a reach of about 894 m. The independent observational check is the coast-to-inland head transect (–28 mm per year), consistent with a moving boundary; the mechanism matches the Whiteford Burrows analogue. This is an accumulating process — it does not settle to an equilibrium.

Climate acts as a spatially uniform decline in the recharge/PET balance, consistent with UKCP18 RCP8.5 projections, lowering the whole water table by the same amount everywhere and drying both slacks across the board. The warren-wide fall is directly observed across the well network; the fitted uniform rate is about –6.35 mm per year ($\approx$ –127 mm over twenty years). It is the pervasive background against which the other three drivers act, and it too accumulates without settling.

Expressed as an equivalent depth averaged over the 949 ha site at a common twenty-year horizon and referenced to the climate shift, the drivers rank as follows:

Driver	Equivalent depth (site-mean, 20 yr)	Relative to climate
Climate (uniform)	−127 mm	100%
Coastal retreat	−78 mm	62%
Dune scrape (net)	−22 mm	17%
Clearfell (relief)	+7 mm	+5%
Broadleaf restock	−4 mm	3%

Table 1: Site-wide footing of each water-table driver. Equivalent depth averaged over the 949 ha site at a common twenty-year horizon, expressed as a percentage of the pervasive climate shift. Of the five, only the coastal drawdown is modelled; the others are observed. (Source: 37b_driver_footing.csv.)

This site-wide ranking differs from the drivers’ local intensity, and the two orderings invert. Coastal retreat has by far the largest *peak* — about −580 mm at the shore over twenty years, several times the climate shift there — but it decays inland over the ~894 m reach; the uniform climate decline is smaller everywhere (−127 mm) but acts across the whole 949 ha, so it carries the largest *site-wide volume*. On the twenty-year reach the coastal curve crosses the flat climate line at about 698 m from the shore (Figure 73, lower panel): coastal dominates shoreward of the crossing, climate inland of it, and because most of the site lies inland the site-mean favours climate. The two statements — coastal strongest at the shore, climate strongest site-wide — are complementary, not in conflict. Of the four footings only the coastal drawdown is modelled; the warren-wide climate fall, the clearfell step and both scrape terms are observed.

The management challenge at Newborough is hydrological adaptation rather than restoration to a prior state — maintaining flooding capacity in an increasingly seasonal climate, while the summer drought trajectory approaches the ecological viability thresholds identified by Curreli et al. (2013) within the next one to two decades.

Four drivers of water-table change at Newborough

schematic · not to scale



Coastal retreat vs climate along the reach inland

water-table curves and 900 m reach to committed scale; dune surface illustrative; vertical exaggerated x62
conceptual reach built from the fitted d0, L and climate background (Section 4.11)

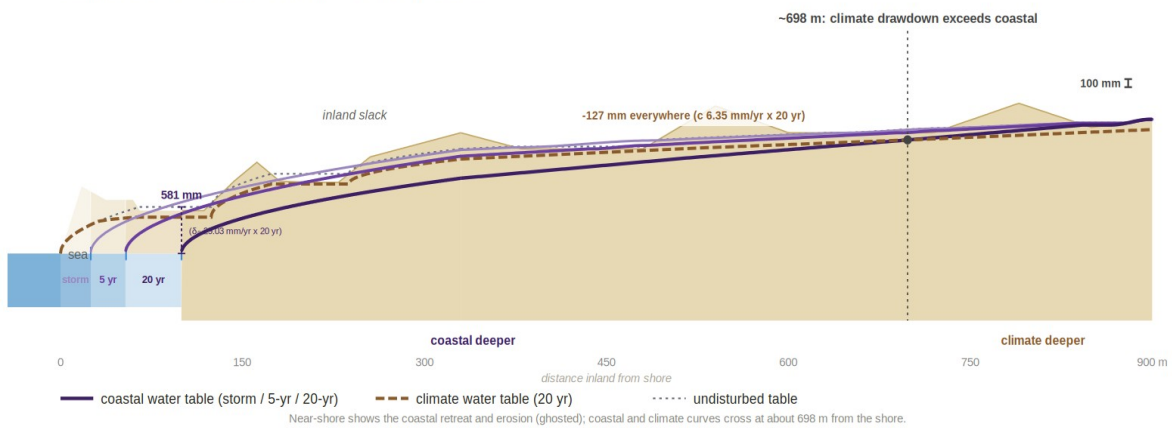


Figure 1.4: Four drivers of the Newborough water table — schematic cross-sections (not to scale; vertical scale exaggerated). Conceptual before/after sections for the site's four water-table drivers — dune scrape and clearfell (local, settling interventions) and coastal retreat and climate (basin-wide, accumulating drivers) — drawn on one shared profile and a common vertical amplitude scale so their relative magnitudes can be read directly. The panels show mechanism and direction, not measured section geometry. Solid line: water table; dashed line: the undisturbed (reference) table. The full-width lower panel places the coastal and climate drawdowns on one continuous reach inland from the shore; they cross at about 698 m, coastal deeper seaward of the crossing and climate deeper inland. Magnitudes, the site-wide footing of each driver and the crossing are developed in the text above. (Source: 09g_mechanism_grid.png.)

1.9 Implications for Restoration and Monitoring

Summer minima are the primary restoration lever. Antecedent summer depth is the dominant predictor of peak winter water table in all western and forest clusters (Section 5.2; Table 12), with winter rainfall amount secondary or non-significant. Restoration success therefore depends on raising summer water table floors through topographic modification and reduced evaporative demand, not on maximizing winter recharge. This reframing has direct implications for monitoring design: summer dipwell readings are more diagnostically informative than winter peak measurements for assessing restoration trajectory in every cluster.

Topographic scraping is the most directly effective intervention identified in the monitoring record, with a measured improvement in summer minimum depth at CEH36 during the pure scraping era representing more than half the gradient separating the SD15b and SD16 community thresholds (Section 5.4). The benefit operates on summer minima and is achievable within the C1/C2 zone where the P_flood threshold remains within reach of mildly wet winters (Section 5.8.1). Scraping steepens the local hydraulic gradient and draws the water table downward upslope (Section 5.4.3), so the spatial planning of any scraping programme must account for the cascade risk and avoid amplifying the existing coastal drainage gradient.

Scraping is therefore not a hydrologically neutral intervention. The same drain-cone physics that quantify the standing forest's drawdown (Section 4.9.3) put an indicative magnitude on the scrape's own neighbour effect: a well 100 m upslope of a CEH36-equivalent scrape sees a modelled drawdown of order -30 mm water-equivalent per month, decaying to roughly -16 at 250 m. These are steady-state modelled figures the 31-month record cannot yet confirm, and they are modest beside the on-site benefit — but they are not zero, and they act in the same direction as, and add to, the coastal-retreat drawdown already deepening the western margin. On the eroding south-western frontage in particular, coastal scraping compounds a drainage gradient the shoreline is already steepening; the intervention is best confined to the mid-eastern C1/C2 transitional zone (Section 5.4.3), where it raises a slack without reinforcing an existing coastal cascade.

The background climate drying trend continues to operate at the scraped site as elsewhere, and the structural benefit at CEH36 has diminished — though not reversed — over the post-felling period. Re-scraping intervals cannot yet be estimated from the available record.

Per-well categorization against the P_flood multiplier (Figure 74) operationalises the priority criterion identified in Conclusion 4. Each well is binned on its λ value into three bands: achievable ($\lambda < 1.5$, reachable in normal-to-mildly-wet winters), marginal ($1.5 \leq \lambda < 2.5$, reachable only in wet winters), and unreachable ($\lambda \geq 2.5$, effectively unreachable under current climate). Across the sixty-five wells in the open-dune clusters C1, C2 and C3, fifty-seven are achievable, eight are marginal, and none are unreachable. By contrast, of the

twenty-three forest-zone wells in C4 and C5, only two sit in the achievable band and four are unreachable — three of the unreachable wells are in C5 Coastal Forest and one in C4 Main Forest. The achievable wells in the C1/C2/C3 transitional zone are the operationally feasible scrape-targeting candidates; the eight marginal open-dune wells define the upper edge of the operational envelope under current climate.

Plantation clearance as implemented in 2017 produced a statistically significant mean water table recovery against the Forest control (Section 4.6.2), confirmed by synthetic extension within the felled compartment (Section 4.6.7). The clearfell signal is concentrated in the mean monthly water level rather than in summer minima, where felled and unfelled wells track each other (Section 5.5).

The evidence does not support the view that the plantation is the principal driver of the warren's summer drying, nor that its removal would reverse it. The metric that governs dune-slack viability — the summer minimum — is deteriorating across felled and unfelled wells alike, driven by a site-wide decline in recharge sensitivity (β_1) that is independent of canopy cover (Section 5.5.1); clearfell raised the mean monthly water level but left the summer minimum unchanged. Where the canopy does shape the climate response, the effect is two-sided rather than simply drying: the standing pine amplifies the wet-year water-table swing through interception-driven recharge suppression (attenuated at the felled well FE2; Section 5.7.5), while the post-felling record shows the felling-attributable rise concentrated in dry conditions (the significant $CWB^2 \times$ clearfell curvature, Section 4.6) — the signature of a canopy that buffered the water table against drought extremes. Removing the forest would thus trade a preliminary dry-season buffering effect and a wet-year amplification for a mean-level gain that does not reach the ecologically binding summer minimum, on the same multi-decadal propagation timescale as the plantation took to develop its influence.

The corrected scenario projections predict small positive net responses at the forest clusters under clearfell and thinning, and near-neutral responses under broadleaf conversion; these magnitudes are substantially smaller than the projected climate-driven Δh range over the same horizon (Section 5.5.2). The spatial framework and BACI record together indicate that forest management perturbations in the C4/C5 zone do not propagate to C1 and C2 where the ecological need is greatest (Sections 5.7, 5.8).

The prediction equations, P_flood thresholds and scenario framework are available as interactive tools hosted on the Newborough Warren Groundwater Research project page (Hollingham, 2026b; https://newbroman.github.io/Newborough_Hydrology/): a per-well flood probability forecaster driven by the P_flood threshold equations, a scenario viewer implementing the SSM equilibrium framework for exploring climate and management scenarios, and a network scatter plot of seasonal extremes referenced against the Curreli et al. (2013) thresholds. The three tools operate on different aspects of the analytical framework and are complementary.

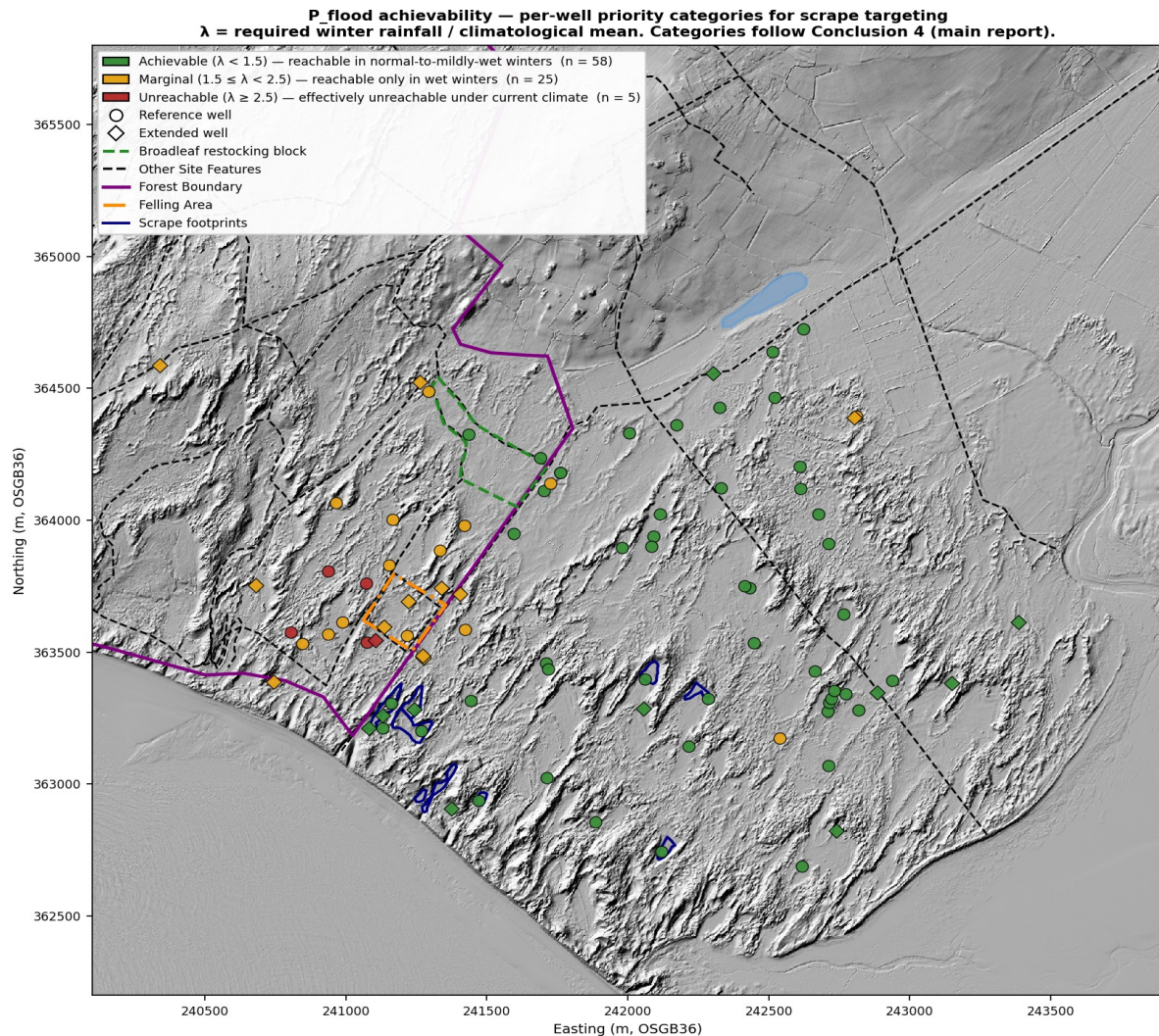


Figure 1.5: Per-well achievability categorization against the P_{flood} rainfall multiplier (λ), the cumulative winter rainfall required to lift each well's summer minimum back above the relevant Curreli et al. (2013) threshold by end of recharge season, expressed as a multiple of climatological winter mean. Wells in the achievable category ($\lambda < 1.5$, green) are reachable in normal-to-mildly-wet winters; marginal wells ($1.5 \leq \lambda < 2.5$, amber) only in wet winters; unreachable wells ($\lambda \geq 2.5$, red) are effectively unreachable under current climate. The cluster pattern (open-dune C1/C2/C3 dominated by achievable; forest C4/C5 dominated by marginal-to-unreachable) operationalises Conclusion 4's priority criterion for scrape-target identification. (Source: 11c_p flood_achievability.png; per-well lookup table in 11c_p flood_achievability_per_well.csv).

Four priorities for future research are identified. First, slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness would enable a calibrated continuous-flow spatial model and directly test the interpretation that substrate thickness, inferred here from elevation and the SSM coefficients, is the dominant control on within-forest coefficient variation (Section 5.6.1). These measurements would also lift the principal limitation on the spatial framework identified in Section 5.7.3, where the absence of calibrated hydraulic conductivity and aquifer thickness data prevents direct testing of cluster-to-cluster propagation.

Second, the C5 Coastal Forest trajectory — the steepest decline in the network in both seasons, with three unresolved candidate explanations (Section 4.8.1; Section 5.7.2) — requires continued monitoring alongside targeted measurement of coastal retreat rates, since coastal boundary retreat is one of the candidate explanations and is the one most directly measurable in the field. Pye and Blott (2024) provide a citable empirical anchor for this priority: dune-toe retreat at the southern Twyni Penrhos frontage of up to 1.8 m yr^{-1} over 2013–2022, projected to continue, with the erosion front migrating northwards toward the western C5 and C3 wells.

Third, the relationship between canopy density and microclimatic shielding could be characterized through monitoring of β_2 responses at wells adjacent to blight-affected stands as the *Dothistroma* needle blight trajectory progresses (Section 5.6.4); this would be an opportunistic rather than designed study, but the natural experiment is already in progress and would require only sustained monitoring at existing or modestly extended sites.

Fourth, the internal structure of the bedrock ridge forming the northern boundary warrants direct investigation. Field observation indicates folds and faults running along the ridge, crossed by step faults, such that parts of it terminate in a stepped fashion — discernible in plan view in the DEM — with one termination immediately north-north-east of CEH14. Such a structure could in principle support an along-ridge sand aquifer, bounded below by the ridge-valley floor, conveying water along the ridge and discharging into the dune system at the structural terminations. The monitoring record does not test this and provides no positive evidence for it: the water balance closes at every cluster without requiring a lateral input (Section 4.9.7), and the diagnostic tests in Supplementary Notes S5 and S6 find neither distance-dependant transport from the ridge nor a residual signature attributable to it. Any such conveyance is therefore either absent, or small and steady enough to be indistinguishable from the fitted coefficients at monthly resolution. Shallow boreholes or geophysical profiling along the ridge, with direct head-gradient measurement at the candidate discharge points, would settle a question the dipwell network is not positioned to answer. It is placed fourth because the hydrological case for it is now the weakest of the four; the structural observation nonetheless stands on its own field evidence, and the northern dune margin's long-term water supply is the interest at stake.